Two-Stage Differences in Differences

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Abstract

This paper develops a framework for estimation and inference to analyze the effect of a policy or treatment in settings with treatment-effect heterogeneity and variation in treatment timing. We propose a two-stage difference-in-differences (2SDD) estimator that compares treated and untreated outcomes after removing group and period effects identified using untreated observations. Our regression-based approach enables us to conduct inference within a conventional GMM asymptotic framework. It easily facilitates extensions such as dynamic treatment effects, triple differences, continuous treatments, time-varying controls, and violations of parallel trends. Simulations of randomly generated placebo laws in state-level wage data demonstrate that 2SDD has rejection rates that are closer to the nominal rate and smaller standard errors than several alternatives in small samples. Across seven empirical applications, we find that 2SDD has a smaller incidence of extreme t-statistics and outlying standard errors.

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1 Introduction

Difference-in-differences (DD) estimation has emerged as an indispensable tool for empirical researchers seeking to evaluate the impact of a given intervention or policy. Its appeal stems in part from the conceptual simplicity of comparing changes in outcomes for groups affected by an intervention to changes for unaffected groups. A potential reason for the widespread use of two-way fixed-effects (TWFE) in settings with multiple groups and time periods is a presumption that it should identify the average effect of the treatment on the treated. Although this intuition is accurate when the heterogeneous treatment effects are distributed identically across groups and periods (a condition that is automatically satisfied in the classic two-group, two-period setting), it does not hold in general. When these distributions are not identical, conditional mean outcomes are no longer linear in group, period, and treatment status, causing the TWFE regression model to be misspecified for conditional mean outcomes, and thus unable to identify the average treatment effect on the treated.

This paper develops a two-stage regression-based approach to identification, along with an accompanying GMM-based variance estimator, which is robust to treatment-effect heterogeneity when adoption of the treatment is staggered over time. The two-stage difference-in-differences (2SDD) estimator, along with valid asymptotic standard errors, can be implemented easily using standard statistical software, with little programming or computational time beyond that required to estimate a regression.[1]

The proposed two-stage methodology regresses outcomes on group and period fixed effects using the subsample of untreated observations in the first stage. The second stage subtracts the estimated group and period effects from observed outcomes and regresses the resulting residualized outcomes on treatment status. Under the usual parallel trends assumption, this procedure identifies the overall average effect of the treatment on the treated (i.e., across groups and periods), even when average treatment effects are heterogeneous over groups and periods. This approach preserves the intuition behind identification in the two-group, two-period case: it recovers the average difference in outcomes between treated and untreated units, after removing group and period effects.

Our approach inherits the flexibility for which researchers have come to appreciate regression as a tool for applied empirical analysis, making it adaptable to the wide variety of settings where difference-in-differences analyses are used in practice, with standard inference procedures naturally applying. This extensibility is aided by our straightforward GMM

[1]We provide example Stata syntax that shows how to implement the two-stage difference-in-differences approach (with valid asymptotic standard errors) via GMM or the `did2s` Stata package (Butts, 2021) in Appendix C; also see the `did2s` R package (Butts and Gardner, 2022).

approach, which naturally and efficiently maintains valid inference within a familiar regression-based framework without requiring additional computational steps. For instance, our basic approach can easily be extended to estimate the dynamic effects of treatments, implement tests of parallel trends, and handle models with individual fixed effects using a variation of the within estimator. It can also accommodate settings where parallel trends holds only after conditioning on time-varying covariates by simply including those covariates as regressors in the first stage, or where covariate-specific trends are needed by interacting time-invariant covariates with time indicators. In settings with continuous treatments, by replacing binary treatment status with a continuous treatment variable in the second stage, our approach preserves interpretability and computational simplicity while ensuring valid inference. Similarly, our method extends seamlessly to triple-differences analyses with only a simple modification to the first stage, offering clear advantages in ease of implementation over alternative methods.

We also present theoretical results that help clarify the relationships among and relative performance of several robust DD estimators, including the imputation estimators proposed by BJS2024 and W2025, which in some cases produce identical point estimates to 2SDD. First, we demonstrate that the choice of parallel trends assumption affects precision: stronger assumptions expand the donor pool of comparison observations, weakly reducing variance by lowering first-stage estimation error. Second, we show that the 2SDD variance is robust to random treatment assignment and remains conservative when treatment assignment is fixed in repeated samples. In contrast, the approach to inference developed in BJS2024, and some variations on those in W2025, rely on fixed treatment status and can be invalid when treatment assignments are stochastic. Third, all three variance estimators can be unified within a common framework, differing in terms of the choice of residual used to construct the variance estimator.

To further evaluate the performance of the proposed variance estimation procedure and the broader methodology, we conduct simulations of randomly generated laws using state-level wage data. This simulation builds on the seminal work of Bertrand, Duflo and Mullainathan (2004), extending their analysis to a setting with heterogeneous treatment effects and staggered treatment timing. In particular, we analyze the performance of various DD estimators and their associated inference procedures with randomly drawn treated states, their associated years of passage, and treatment effects. Using a 42-year panel, we compare rejection rates at the 5 percent significance level from recently proposed heterogeneity-robust procedures (CS2021; SA2021; W2025; BJS2024; dCDH2024).[2] Simulations highlight the value of our

[2]Our analysis does not include the local projections approach (Dube et al., 2023) due to its lack of a theoretical framework for inference.

approach to estimation and inference by demonstrating its finite-sample performance. Our procedure achieves well-calibrated rejection rates and favorable precision while retaining the speed of imputation-style estimation. This combination is particularly useful in comparison with the imputation approach from BJS2024, which provides identical point estimates to ours but uses different variance estimators.[3]

We also illustrate the practical implications of the different inference procedures using data from Lafortune, Rothstein and Schanzenbach (2018) on school finance reforms. This application closely mirrors the finite-sample issues highlighted in the simulations. Because the BJS2024 and W2025 estimators share the same point estimates, we focus on comparing alternative variance estimators while holding the estimated treatment effects fixed. We use other estimators primarily as benchmarks to help interpret the resulting differences in inference. We then place this example in the context of a broader replication exercise covering seven empirical applications (Supp. Appendix A). Across these applications, 2SDD tends to produce comparatively stable standard errors and t-statistics, while other approaches generate more frequent outliers or inconsistencies relative to alternative approaches.

Our work adds to an emerging body of research highlighting limitations of the traditional TWFE approach for DD estimation in the presence of staggered treatment timing and when the effects of a treatment vary across groups and time.[4] We motivate our approach by elucidating how misspecified TWFE regression models project heterogeneous treatment effects onto treatment status, group effects, and period fixed effects. The simple observation that untreated outcomes are linear in group and period effects under parallel trends then naturally leads to our proposed two-stage method. Several papers provide alternative representations of the TWFE estimand. BJS2024 show that TWFE identifies a regression-weighted mean of the average effect of the treatment in each post-treatment period, and dCDH2020 show that all TWFE regression estimates (which include DD regressions as a special case) identify weighted averages of group- and period-specific average treatment effects. Since the weights in both of these representations can be negative, interpreting the TWFE estimand becomes challenging. Goodman-Bacon (2021) further shows that the TWFE estimate represents a weighted average

[3]The "imputation" estimator, which first appears in Borusyak, Jaravel and Spiess (2021), is numerically identical to the two-stage estimators initially proposed by Gardner (2020), Thakral and Tô (2020), and Liu, Wang and Xu (2019). However, they develop a different asymptotic theory, resulting in an asymptotically conservative default variance estimator and a leave-one-out modification which they show results in improved finite-sample performance. BJS2024 note that similar approaches have been proposed for factor models by Gobillon and Magnac (2016) and Xu (2017), while Arkhangelsky and Imbens (2024) note that the two-stage approach can be viewed as a particular extension of a broader class of estimation strategies for panel models developed by Chamberlain (1992) and extended by Graham and Powell (2012) and Arellano and Bonhomme (2012).

[4]See, for example, dCDH2020, Goodman-Bacon (2021), Imai and Kim (2021), SA2021, Athey and Imbens (2022), and BJS2024.

of all two-group, two-period differences in differences, which under parallel trends identifies a combination of weighted averages of group×period-specific average treatment effects and changes over time in those effects. These decomposition results tend to motivate alternative methodologies based on manually averaging cohort×period-specific average treatment effects (dCDH2020; CS2021; SA2021).[5] Although Harmon (2024) shows how these procedures are more efficient than imputation-based procedures such as BJS2024 when residuals are strongly serially correlated, the simplicity of our procedure makes it easy to adapt to be more efficient in the presence of serial correlation.

In the presence of staggered treatment adoption, several alternatives to the TWFE regression approach exhibit robustness to heterogeneity across groups and periods. One alternative, as mentioned earlier, is to estimate separate average treatment effects for each group and period, which can then be aggregated to form measures of the overall effect of the treatment. In comparison to this approach, our regression-based methodology offers simplicity in estimation and inference, significant computational speed advantages, and strong finite-sample performance. In addition, our approach retains the efficiency advantages pointed out by BJS2024. We also discuss how to mitigate bias from violations of parallel trends by using an appropriate subset of untreated observations in the first stage.

Alternative regression-based approaches include the "stacked" difference-in-differences (see, e.g., Gormley and Matsa, 2011; Deshpande and Li, 2019; Cengiz et al., 2019; Dube et al., 2023), which attempts to transform the staggered adoption setting to a two-group, two-period design (in which difference in differences identifies the overall average effect of the treatment on the treated) by stacking separate datasets containing observations on treated and control units for each treatment cohort, and the extended TWFE approach (W2025). Several limitations arise when applying these methods. For example, as we show in our supplemental material, the stacked estimator identifies a particular weighted average of group-specific average treatment effects that depends on arbitrary features of the data, making the resulting estimate more challenging to interpret.[6] Implementing the stacked approach also requires defining a fixed event time window and ensuring a balanced panel throughout this period, and involves using the same control groups across different stacked datasets without a theoretical framework for inference. Relative to these approaches, ours delivers clear and interpretable estimates, provides a straightforward theoretical framework for inference, and allows for flexible implementation across various contexts, including those with time-varying covariates that interact arbitrarily with treatment effects and the other

[5] Gibbons, Suárez Serrato and Urbancic (2018) also suggest an approach like this for fixed effects models.

[6] The weights depend on the relative sizes of the group-specific datasets and the variance of treatment status within those datasets, as Appendix D.2 shows.

extensions of our procedure discussed above.

Given the multitude of alternative approaches for DD estimation, our empirical and simulation exercises constitute a distinct contribution to this literature. Our empirical exercises complement recent work by Chiu et al. (2023), which reanalyzes a set of political science publications that estimate TWFE regressions. They emphasize that TWFE estimates correlate strongly with the estimates from alternative methods but find that the latter tend to be less precise. Under homogeneous treatment effects, as our simulation results demonstrate, our proposed approach to inference achieves the closest standard error to that of a TWFE estimator that imposes a null effect in the pre-treatment periods.[7] Our simulation exercises present a novel and systematic comparison of standard errors and rejection rates across different estimators in staggered adoption settings based on typical empirical applications. Subsequent work by Egerod and Hollenbach (2024) and Weiss (2024) also compare heterogeneity-robust estimators using simulations. Weiss (2024) focuses on the CS2021, SA2021, BJS2024, and dCDH2024 approaches (but does not examine 2SDD) and shows that their built-in variance estimators can perform poorly in small samples. Egerod and Hollenbach (2024) reinforce our finding of systematically low coverage among most methods, with CS2021 and 2SDD providing significantly better coverage. They further document power issues that arise in small samples and show that BJS2024 exhibits a notably high probability of reporting the wrong sign for the estimated effect conditional on a significant result. As our method for achieving robustness to treatment-effect heterogeneity entails minimal efficiency loss in settings where TWFE provides an unbiased estimate, it offers arguably the most compelling alternative to the TWFE approach in practice.

The paper proceeds as follows. In Section 2, we provide intuition for why the TWFE approach to DD estimation may not identify the average effect of the treatment on the treated, and show how our proposed two-stage regression-based approach is robust to treatment-effect heterogeneity in settings with variation in treatment timing. Section 3 contains theoretical results about the relationship between 2SDD and alternative approaches. Sections 4 and 5 demonstrate the performance of the two-stage approach compared to alternative proposals in simulations and empirical applications. We conclude in Section 6.

[7] As a result, our proposed estimator results in greater precision than the fully dynamic event-study specification using TWFE.

2 Inference with a two-stage approach

2.1 Setting

Let $i \in \{1, \dots, N\}$ index units (e.g., states or, with microdata, individuals) and $t \in \{1, \dots, T\}$ index calendar time (often years). Let $\Omega = \{(i,t) : i = 1, \dots, N,\ t = 1, \dots, T\}$ denote the full panel, and partition it into the set of treated observations $\Omega_1 = \{(i,t) : D_{it} = 1\}$ and untreated observations $\Omega_0 = \{(i,t) : D_{it} = 0\}$, where $D_{it} \in \{0,1\}$ indicates whether unit i is treated at time t.

Groups and periods. Partition units and time into treatment groups $g \in \{1, \dots, G, \infty\}$ and periods $p \in \{0, 1, \dots, P\}$ defined by the adoption of the treatment among successive groups, so that members of group ∞ are untreated in all periods, only members of group 1 are treated in period 1, members of groups 1 and 2 are treated in period 2, and so on. Periods $p \in \{0, 1, ..., P\}$ coarsen calendar time t by grouping dates according to which successive treatment cohort first becomes active. Let G_i and P_t denote the corresponding group and period variables. We assume for simplicity that the treatment is both irreversible and unanticipated (though these assumptions can be at least partially relaxed, as detailed in Appendix F).

Potential outcomes and treatment effects. Let Y_{it} denote the observed outcome for unit i at time t, and let $Y_{it}(0)$ denote the potential outcome that would be obtained if the unit had never been treated. The treatment effect for individual i at time t is

$$\beta_{it} \ := \ Y_{it} - Y_{it}(0),$$

which is by construction stochastic and heterogeneous across units and time.[8] The group×period average treatment effect is

$$\beta_{gp} \ := \ \mathbb{E}\big[Y_{it} - Y_{it}(0) \mid G_i = g,\ P_t = p\big].$$

Our baseline target object is the *average treatment effect on the treated* (ATT),

$$\beta \ := \ \mathbb{E}\big[\beta_{it} \mid D_{it} = 1\big],$$

[8]Treatment effects for the never-treated observations may be normalized to zero, since they are not identified.

where the expectation is taken over all treated observations.[9] We implicitly assume that this object is well-defined in that $\mathbb{E}[D_{it}] > 0$.

Observed data and covariates. We observe (Y_{it}, X_{it}, D_{it}), where X_{it} denotes a K-dimensional vector of covariates (which may include time indicators and are assumed to be unaffected by treatment status).[10] As in Callaway and Sant'Anna (2021), Sun and Abraham (2021), and Wooldridge (2025), we assume that $\{Y_{it}(0), Y_{it}, D_{it}, X_{it}\}_{t=1}^{T}$ are i.i.d. across individuals.[11] We also assume that T is fixed and $N \to \infty$, the most common case encountered in practice.

2.2 The problem with difference-in-differences regression

In this subsection, we motivate the challenges in difference-in-differences (DD) designs in the case without covariates. DD research designs attempt to identify the causal effects of treatments under the parallel or common trends assumption. This assumption asserts that, absent the treatment, treated units would experience the same change in outcomes as untreated units. Mathematically, this amounts to the assumption that average untreated potential outcomes decompose into additive group and period effects. Under parallel trends, mean outcomes satisfy

$$\mathbb{E}[Y_{it} \mid g, p, D_{it}] = \lambda_g + \alpha_p + \beta_{gp} D_{it}. \tag{1}$$

The idea behind DD is to eliminate the permanent group effects λ_g and secular period effects α_p in order to identify the average effect of the treatment. In the classic setup, there are only two periods (pre and post) and two groups (treatment and control). In this setting, within-group differences over time eliminate the group effects and within-period differences between groups eliminate the period effects. Hence the between-group difference in post-pre differences (i.e., the difference in differences) identifies the average effect of the treatment for members of the treatment group during the post-treatment period.

[9]This definition implicitly treats the treatment effects β_{it} as a single random variable whose distribution varies with time, rather than a sequence of time-indexed random variables. Thinking of the treatment effects this way, $\mathbb{E}[\beta_{it}|D_{it} = 1] = \mathbb{E}[\beta_{it} D_{it}]/\mathbb{E}[D_{it}] = [\sum_t \mathbb{E}(\beta_{it} D_{it}|t)]/[\sum_t \mathbb{E}(D_{it}|t)]$. Alternatively, we could simply define the estimand to be $[\sum_t \mathbb{E}(\beta_{it} D_{it})]/[\sum_t \mathbb{E}(D_{it})]$. One advantage of this former approach is that it clarifies the weights with which our estimator aggregates group- and period-specific treatment effects, since regardless of whether covariates are included in the first stage, we have that $\mathbb{E}(\beta_{it}|D_{it} = 1) = \sum_{g,p} \beta_{gp} \Pr(g, p)$.

[10]This assumption is actually stronger than necessary. As Callaway, Goodman-Bacon and Sant'Anna (2021) note, it suffices to assume that the treated and untreated covariate distributions are the same. Nonetheless, it is comparable to Assumptions 4 and 5 in CS2021.

[11]This i.i.d. setup is somewhat stronger than merely having clustered error terms, but our derivation will still go through even if the observations are i.n.i.d.

The two-period, two-group difference-in-differences estimate can be obtained using a regression of outcomes on group and period fixed effects and a treatment-status indicator:

$$Y_{it} = \lambda_{g(i)} + \alpha_{p(t)} + \beta D_{it} + \varepsilon_{it}, \tag{2}$$

where $g(i)$ is the group index for unit i and $p(t)$ is the period index at time t. It follows from (1) that the coefficient on D_{it} in (2) identifies the average effect of the treatment on the treated, $\mathbb{E}[Y_{it} - Y_{it}(0) \mid D_{it} = 1]$.[12]

The regression approach suggests a natural way to extend the DD idea to settings with multiple groups and time periods. Unfortunately, as several authors have noted (dCDH2020; Goodman-Bacon, 2021; Imai and Kim, 2021; Athey and Imbens, 2022; BJS2024), when the average effect of the treatment varies across groups and over periods, the coefficient on D_{it} in specification (2) does not always identify an easily interpretable measure of the "typical" effect of the treatment. Although this result is now well established, because it is also somewhat counterintuitive, it bears further clarification.

While there are multiple ways to think about the typical effect of the treatment when that effect varies across groups and over time, an obvious candidate is the average $\beta :=$ $\mathbb{E}[Y_{it} - Y_{it}(0) \mid D_{it} = 1]$ of group- and period-specific average treatment effects, taken over all units that receive the treatment and all times during which they receive it. Hence, parallel trends can be expressed as

$$\mathbb{E}[Y_{it} \mid g, p, D_{it}] = \lambda_g + \alpha_p + \beta D_{it} + (\beta_{gp} - \beta) D_{it}.$$

The difficulty with the regression approach is that, except in special cases, the "error term" $(\beta_{gp} - \beta) D_{it}$ in this expression varies at the group$\times$period level, and is not mean zero *conditional on group membership, period, and treatment status.* Consequently, the regression is misspecified in the sense that the conditional expectation $\mathbb{E}[Y_{it} \mid g, p, D_{it}]$ is not a linear function of those variables (at least, not one in which the coefficient on D_{it} is β). In contrast to the two-group, two-period case, the coefficient on D_{it} from the regression DD specification (2) does not identify β unless $\beta_{gp} = \beta$. Outside of this special case, when average treatment effects vary across groups and periods, and the adoption of the treatment by different

[12]There are several equivalent variations on this regression. Specification (2) is identical to a regression of outcomes on an indicator $Post_{it}$ for whether t occurs in the post-treatment period, an indicator $Treat_{it}$ for whether i belongs to the treatment group, and an interaction between the two. Often, the group and period effects λ_g and α_p in (2) are replaced with individual and time effects λ_i and γ_t. By the Frisch-Waugh-Lovell theorem, the coefficient on D_{it} in (2) can be obtained by regressing Y_{it} on the residuals from a regression of treatment status on group and period effects. Since treatment status only varies by group and period, these residuals are the same as those from a regression of treatment status on individual and time effects, so the coefficients on treatment status from both specifications are identical.

groups is staggered over time, difference-in-differences regression does not recover a simple group×period average treatment effect (dCDH2020; Goodman-Bacon, 2021; BJS2024). In Appendix D.1, we derive a simple expression for what regression DD does identify in the special case of cohort fixed effects and no covariates.

2.3 A two-stage approach

The observation that the problem with traditional TWFE approaches arises from misspecification of Equation (2) suggests a simple two-stage average treatment effect estimator for the multiple group and period case. As long as there are untreated and treated observations for each group and period, λ_g and α_p are identified from the subpopulation of untreated groups and periods. The overall group×period average effect of the treatment on the treated is then identified from a comparison of mean outcomes between treated and untreated groups, after removing the group and period effects.

This logic suggests the following regression-based two-stage estimation procedure:

1. Estimate the model
$$Y_{it} = \lambda_{g(i)} + \alpha_{p(t)} + u_{it}$$
on the sample of observations for which $D_{it} = 0$, retaining the estimated group and time effects $\hat{\lambda}_g$ and $\hat{\alpha}_p$.
2. Regress adjusted outcomes $Y_{it} - \hat{\lambda}_{g(i)} - \hat{\alpha}_{p(t)}$ on D_{it}.

Since parallel trends implies that

$$\mathbb{E}[Y_{it} \mid g, p, D_{it}] - \lambda_g - \alpha_p = \beta_{gp} D_{it} = \beta D_{it} + (\beta_{gp} - \beta) D_{it},$$

where $\mathbb{E}[(\beta_{gp} - \beta) D_{it} \mid D_{it}] = 0$, this procedure identifies β, even when the adoption and average effects of the treatment are heterogeneous with respect to groups and periods. Observe that the expression $\mathbb{E}[(\beta_{gp} - \beta) D_{it} \mid D_{it}] = 0$ no longer conditions on g and p, which is possible as we only regress on D_{it} (without group and period indicators) in the second stage. A straightforward application of the continuous mapping theorem shows that this two-stage estimator is consistent for the overall average effect of the treatment.

We now formalize this argument and extend it to specifications that include time-varying covariates and unit-level fixed effects. Our difference-in-differences approach is based on the following parallel trends assumption.

Assumption 1 (Parallel trends)**.** *There exist non-stochastic γ, λ_i such that, for all it,*

$$\mathbb{E}\left[Y_{it}(0) \mid \{X_{it}\}_{t=1}^{T}, G_i = g\right] = \lambda_i + X_{it}'\gamma. \tag{3}$$

This notation subsumes the familiar time fixed effects into the vector X_{it} as time indicators.[13] This definition of parallel trends mimics Assumption 1 of BJS2024 who treat G_i and X_{it} as non-stochastic objects. Note that conditioning on the treatment group g is equivalent to conditioning on a particular sequence $\{D_{it}\}_{t=1}^{T}$ of treatment paths.

Under Assumption 1, we can express observed outcomes using the regression specification

$$Y_{it} = \lambda_i + \beta D_{it} + X_{it}'\gamma + \varepsilon_{it}, \tag{4}$$

where ε_{it} is defined as the regression error, and the observed outcomes are given by $Y_{it} = Y_{it}(0) + \beta_{it} D_{it}$. To remove the individual fixed effects, we define the transformation $\tilde{Y}_{it} := Y_{it} - \frac{1}{T_i^0}\sum_{t=1}^{T_i^0} Y_{it}$, where $T_i^0 := \sum_{t=1}^{T}(1 - D_{it})$ is the last time period before individual i gets treated (or equivalently, the number of periods in which i is observed in an untreated state). For $\tilde{Y}_{it}$ to be well-defined, we assume that, for all i, there is some t where $D_{it} = 0$. The transformed $\tilde{X}_{it}$ and $\tilde{\varepsilon}_{it}$ are defined in a similar manner.

Using this transformation, (4) can also be written as

$$\tilde{Y}_{it} = \beta D_{it} + \tilde{X}_{it}'\gamma + \tilde{\varepsilon}_{it}.$$

Lemma 1. *If Assumption 1 holds, then* $\mathbb{E}\left[\sum_t D_{it}\tilde{\varepsilon}_{it}\right] = 0$ *and* $\mathbb{E}\left[\sum_t \tilde{X}_{it}\tilde{\varepsilon}_{it} \mid D_{it} = 0\right] = 0$.

Assumption 1 is hence the key identifying assumption for this procedure, as it implies moment conditions for regressions. Due to Lemma 1, running a standard regression of $\tilde{Y}_{it}$ on $\tilde{X}_{it}$ for observations with $D_{it} = 0$ yields a consistent estimator for γ. We refer to this regression that obtains γ as the first-stage regression. Then, using the estimated $\hat{\gamma}$, we can regress $\tilde{Y}_{it} - \tilde{X}_{it}\hat{\gamma}$ on D_{it} to obtain an estimate of β; we refer to this regression as the second-stage regression.

To be precise about these regressions, we define a few objects. Using X_{kit} to denote regressor k for individual i at time t, we define the $T \times K$ matrix $\tilde{X}_{0i}$ as

$$\tilde{X}_{0i} = \begin{bmatrix} \left(X_{1i1} - \frac{1}{T_i^0}\sum_{t=1}^{T} X_{1it}(1 - D_{it})\right)(1 - D_{i1}) & \cdots & \left(X_{Ki1} - \frac{1}{T_i^0}\sum_{t=1}^{T} X_{Kit}(1 - D_{it})\right)(1 - D_{i1}) \\ \vdots & \ddots & \vdots \\ \left(X_{1iT} - \frac{1}{T_i^0}\sum_{t=1}^{T} X_{1it}(1 - D_{it})\right)(1 - D_{iT}) & \cdots & \left(X_{KiT} - \frac{1}{T_i^0}\sum_{t=1}^{T} X_{Kit}(1 - D_{it})\right)(1 - D_{iT}) \end{bmatrix}.$$

[13] If X_{it} consists of a set of time indicators, this assumption takes form $\mathbb{E}[Y_{it}(0) \mid G_i = g] = \lambda_i + \alpha_t$, where α_t are time fixed effects

Similarly, $\tilde{Y}_{0i}$ is a $T \times 1$ vector of the form

$$\tilde{Y}_{0i} = \left[\tilde{Y}_{i1}\left(1 - D_{i1}\right) \quad \cdots \quad \tilde{Y}_{iT}\left(1 - D_{iT}\right)\right]'.$$

The coefficient estimator from the first stage regression is then

$$\hat{\gamma} = \left(\sum_{i=1}^{N} \tilde{X}_{0i}' \tilde{X}_{0i}\right)^{-1} \left(\sum_{i=1}^{N} \tilde{X}_{0i}' \tilde{Y}_{0i}\right).$$

Observe that both sums are over independent individuals i (the sum over time is already implicit in the matrix multiplication) and that the second-stage regression is done without a constant, because the data are already demeaned. Hence, the two-stage difference-in-differences estimator is

$$\hat{\beta} = \left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\right)^{-1} \left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\left(\tilde{Y}_{it} - \tilde{X}_{it}\hat{\gamma}\right)\right).$$

To summarize, the two-stage procedure is:

1. Regress $\tilde{Y}_{0i}$ on $\tilde{X}_{0i}$ to obtain $\hat{\gamma}$.
2. Regress adjusted outcomes $\tilde{Y}_{it} - \tilde{X}_{it}'\hat{\gamma}$ on D_{it} to obtain $\hat{\beta}$.

For these regressions to be feasible, we need to assume a rank condition such that $\left(\sum_{i=1}^{N} \tilde{X}_{0i}' \tilde{X}_{0i}\right)$ is invertible. This invertibility condition rules out identification of unit and time fixed effects separately in environments where treatment cohorts are too small and we are using too few periods.

Assumption 2 (Rank condition)**.** $\mathbb{E}[\tilde{X}_{0i}' \tilde{X}_{0i}]$ *and* $\mathbb{E}[D_{it}]$ *are invertible.*

Inference can then proceed by standard GMM arguments. To apply limit theorems, it suffices to assume that the moments of stochastic objects are bounded.

Assumption 3 (Bounded moments)**.** *There exists* $C < \infty$ *such that* $\mathbb{E}[Y_{it}(0)^4] \leq C$, $\mathbb{E}[X_{kit}^8] \leq C$, *and* $\mathbb{E}[\beta_{it}^4] \leq C$ *for all* t.

The first- and second-stage regressions can be written as the solution to the sample analog of the following moment conditions:

$$\mathbb{E}\left[\begin{array}{c} \tilde{X}_{0i}'\left(\tilde{Y}_{0i} - \tilde{X}_{0i}\gamma\right) \\ \sum_{t=1}^{T} D_{it}\left(\tilde{Y}_{it} - \tilde{X}_{it}'\gamma - \beta D_{it}\right) \end{array}\right] = 0,$$

where we have $K+1$ moment conditions, with K in the first stage and one in the second stage.[14]

Theorem 1. *If Assumptions 1 to 3 hold, then* $\hat{\gamma} \xrightarrow{p} \gamma$, $\hat{\beta} \xrightarrow{p} \beta$ *and* $\sqrt{N}\left(\hat{\beta}-\beta\right) \xrightarrow{d} N(0,V)$, *where* $V = G_\beta^{-1}\mathbb{E}\left[\left(g_i + G_\gamma \psi_i\right)\left(g_i + G_\gamma \psi_i\right)'\right] G_\beta^{-1\prime}$, *with*

$$
\begin{aligned}
G_\beta &= -\mathbb{E}\left[\sum_{t=1}^{T} D_{it}\right], \\
G_\gamma &= -\mathbb{E}\left[\sum_{t=1}^{T} D_{it}\tilde{X}_{it}\right], \\
\psi_i &= \mathbb{E}\left[\tilde{X}_{0i}'\tilde{X}_{0i}\right]^{-1}\tilde{X}_{0i}'\left(\tilde{Y}_{0i} - \tilde{X}_{0i}\gamma\right),
\end{aligned}
$$

and

$$
g_i = \sum_{t=1}^{T} D_{it}\left(\tilde{Y}_{it} - \tilde{X}_{it}'\gamma - \beta D_{it}\right)
$$

The sample analogs are:

$$
\begin{aligned}
\hat{G}_\beta &= -\frac{1}{N}\sum_i \sum_t D_{it}, \\
\hat{G}_\gamma &= -\frac{1}{N}\sum_i \sum_t D_{it}\tilde{X}_{it}, \\
\hat{\psi}_i &= \left[\frac{1}{N}\sum_i \tilde{X}_{0i}'\tilde{X}_{0i}\right]^{-1}\left(\tilde{X}_{0i}'\left(\tilde{Y}_{0i} - \tilde{X}_{0i}\hat{\gamma}\right)\right),
\end{aligned}
$$

and

$$
\hat{g}_i = \sum_t D_{it}\left(\tilde{Y}_{it} - \tilde{X}_{it}'\hat{\gamma} - \hat{\beta} D_{it}\right).
$$

With $\hat{V} = \hat{G}_\beta^{-1}\frac{1}{N}\sum_i\left[\left(\hat{g}_i + \hat{G}_\gamma\hat{\psi}_i\right)\left(\hat{g}_i + \hat{G}_\gamma\hat{\psi}_i\right)'\right]\hat{G}_\beta^{-1}$, *we have* $\hat{V} \xrightarrow{p} V$.

Theorem 1 states that the 2SDD estimator is consistent for β, and is asymptotically normal. Further, the plug-in variance estimator is consistent, which suffices for feasible inference. The proof in Appendix A proceeds by standard arguments (see Newey and McFadden 1994).

[14]Note that the joint solution to this GMM problem is numerically identical to the "manual" two-stage procedure described above.

2.4 Event studies

This procedure can easily be extended to event-study regressions. DD analyses are often accompanied by event-study regressions of the form

$$Y_{it} = \lambda_{g(i)} + \alpha_t + \sum_{r=-\underline{R}}^{\overline{R}} \eta_r W_{rit} + u_{it}, \tag{5}$$

where for $r \leq 0$ the $W_{rit} \in \{W_{-\underline{R}it}, \dots, W_{0it}\}$ are $(r+1)$-period leads of treatment adoption, and for $r > 0$ the $W_{rit} \in \{W_{1it}, \dots, W_{\overline{R}it}\}$ are r-period lags of adoption (i.e., $W_{rit} = 1[t - t^*(i) = r]$, where $t^*(i)$ denotes the time at which individual i becomes treated). SA2021 show that, when duration-specific average treatment effects vary across groups, event-study regressions suffer from the same problem as DD regressions.[15]

The two-stage procedure developed above can be extended to the event-study setting by amending the second stage of the procedure to:

2'. Regress $Y_{it} - \hat{\lambda}_{g(i)} - \hat{\alpha}_t$ on $W_{-\underline{R}it}, \dots, W_{0it}, \dots, W_{\overline{R}it}$.

Following the logic of the previous section, because $\mathbb{E}[Y_{it} \mid g, t, (W_{rit})] - \lambda_g - \alpha_t$ is linear in the W_{rit}, the coefficients on the W_{rit}, $r > 0$, identify the average effects $\mathbb{E}[\beta_{it} \mid W_{rit} = 1]$.[16] For $r \leq 0$, the coefficients on the W_{rit} can be used to test the hypothesis that $\mathbb{E}[Y_{it} \mid g, t, W_{rit} = 1] = \lambda_g + \alpha_t$ (i.e., that the mean first-stage population residual is zero for all units who are $r + 1$ periods away from adopting the treatment), as implied by parallel trends. We provide additional discussion of testing parallel trends within this framework in Appendix E. Note that, by the same logic, the treatment-duration indicators in step 2' can be replaced with group- or period-specific treatment-status indicators to identify group- or period-specific ATTs.

To formalize and extend this argument, we assume that there is an $\underline{R}^*$ such that parallel trends holds for all time periods $t - t^*(i) < -\underline{R}^*$. Researchers typically assume that $\underline{R}^* = 0$, so that parallel trends holds in all pre-treatment periods. Allowing $\underline{R}^* > 0$ allows us to relax the assumption that the treatment is unanticipated (and provides us with another method of testing whether parallel trends is satisfied).

[15] An argument similar to the one presented for DD regressions in Section 2.2 can also be used to show that the coefficients on the W_{rit} do not identify the average effect of being treated for r periods.

[16] This expectation is taken over all groups with treatment durations of at least r. Since under staggered adoption the completed treatment duration varies by group, the groups over which these duration-specific effects are averaged will vary across durations. These averages are also what the interaction-weighted estimator proposed by SA2021 identifies. If all groups are treated for at least $\bar{P}$ periods, an alternative is to exclude observations corresponding to treatment durations longer than $\bar{P}$ periods from the second-stage sample, in which case the two-stage approach identifies duration-specific treatment effects, averaged over all groups.

The W_{rit} can be stacked into an $\underline{R} + \overline{R} + 1$-dimensional vector W_{it}. Define $Q_{it} := 1\left[t - t^*(i) < -\underline{R}^*\right]$. Then, by analogy to the case for the overall ATT, define $T_i^Q := \sum_{t=1}^{T} Q_{it}$. Now,

$$\tilde{Y}_{it} = Y_{it} - \frac{1}{T_i^Q} \sum_{t=1}^{T_i^Q} Y_{it} Q_{it},$$

with a similar definition applying to $\tilde{X}_{it}$ and $\tilde{\varepsilon}_{it}$. Analogously,

$$\tilde{X}_{Qi} = \left[\begin{array}{ccc} \left(X_{1i1} - \frac{1}{T_i^Q} \sum_{t=1}^{T} X_{1it} Q_{it}\right) Q_{i1} & \cdots & \left(X_{Ki1} - \frac{1}{T_i^Q} \sum_{t=1}^{T} X_{Kit} Q_{it}\right) Q_{i1} \\ \vdots & & \\ \left(X_{1iT} - \frac{1}{T_i^Q} \sum_{t=1}^{T} X_{1it} Q_{it}\right) Q_{iT} & \cdots & \left(X_{KiT} - \frac{1}{T_i^Q} \sum_{t=1}^{T} X_{Kit} Q_{it}\right) Q_{iT} \end{array} \right],$$

and

$$\tilde{Y}_{Qi} = \left[\tilde{Y}_{i1} Q_{i1} \quad \cdots \quad \tilde{Y}_{iT} Q_{iT}\right].$$

In this environment, our analogous two-stage procedure becomes:

1. Regress $\tilde{Y}_{Qi}$ on $\tilde{X}_{Qi}$ to obtain $\hat{\gamma}$.

2. Regress adjusted outcomes $\tilde{Y}_{it} - \tilde{X}_{it}'\hat{\gamma}$ on W_{it} to obtain $\hat{\eta}$.

Hence, the first- and second-stage estimators are

$$\hat{\gamma} = \left(\frac{1}{N} \sum_i \tilde{X}_{Qi}' \tilde{X}_{Qi}\right)^{-1} \left(\frac{1}{N} \sum_i \tilde{X}_{Qi}' \tilde{Y}_{Qi}\right),$$

and

$$\hat{\eta} = \left(\sum_{i=1}^{N} \sum_{t=1}^{T} W_{it} W_{it}'\right)^{-1} \left(\sum_{i=1}^{N} \sum_{t=1}^{T} W_{it} \left(\tilde{Y}_{it} - \tilde{X}_{it} \hat{\gamma}\right)\right).$$

As before, the estimators can be written as the solution to a GMM problem:

$$\mathbb{E} \left[\begin{array}{c} \tilde{X}_{Qi}' \left(\tilde{Y}_{Qi} - \tilde{X}_{Qi} \gamma\right) \\ \sum_{t=1}^{T} W_{it} \left(\tilde{Y}_{it} - \tilde{X}_{it}' \gamma - W_{it}' \eta\right) \end{array} \right] = 0.$$

The normality and consistency results are analogous.

Remark 1. Our method's conceptual simplicity, and reliance on the method of moments for inference, make it easy to extend in several ways. While we explain the extension to event studies in some detail in this subsection, our method can be extended among other dimensions as well. These include group by time specific ATTs, weaker parallel trend assumptions, triple differences, standard errors with serial correlation, continuous and multivalued treatments, anticipation effects, reversible treatments, design-based analysis, linear combinations of treatment effects, and to testing for treatment effect heterogeneity, which are explained in Appendix F.

3 Theoretical Results on Comparative Variances and Variance Estimators

This section develops theoretical results that clarify the properties of the 2SDD estimator relative to the alternative approaches that can produce identical point estimates. Section 3.1 shows that a stronger parallel trends assumption, by expanding the valid donor pool, weakly reduces the variance of the common imputation estimator. Section 3.2 shows how the 2SDD approach to inference is exact when treatment status is random but conservative when treatment status is fixed in repeated samples, while the way that the variances are constructed in BJS2024 and W2025 may be anti-conservative under random treatment. Section 3.3 then compares the three target variance estimands and provides conditions where they coincide.

Throughout, the common imputation estimator is written as

$$\hat{\beta} \;=\; w'(Y_1 - X_1\hat{\gamma}), \qquad \hat{\gamma} = (X_0'X_0)^{-1}X_0'Y_0, \tag{6}$$

where Y_1 and X_1 stack observations in the set of treated observations $\Omega_1(D)$ (where D is the $N \times T$ vector of treatment status realizations for each observation), Y_0 and X_0 stack observations in the donor pool $\Omega_0(D)$ of untreated observations, $w \in \mathbb{R}^{N_1}$ is a pre-specified aggregation weight vector with $\mathbf{1}'w = 1$, $X_{it} \in \mathbb{R}^K$ collects nuisance regressors (e.g., unit and time indicators), and $c := X_1'w$ summarises the interaction between nuisance regressors and aggregation weights.

We note that 2SDD, BJS2024, and the W2025 ETWFE (extended two-way fixed effects) regression are algebraically identical as point estimators under a common implementation.[17] As this result is already stated in Wooldridge (2025), we state it here for completeness and defer the derivation to Appendix B.1.

[17]Without covariates, all three estimators produce the same point estimates. With covariates, the ETWFE estimator produces different point estimates because it is based on a specification that includes interactions between covariates and cohort and time indicators (however, this specification can be modified to produce estimates that coincide with 2SDD and imputation approaches).

3.1 Impact of Parallel Trends Assumptions

The parallel trends assumption governs which untreated observations form a valid donor pool $\Omega_0(D)$ for the first-stage regression. A stronger assumption expands the donor pool; a weaker one restricts it. We show that expanding the donor pool weakly reduces the variance of the common imputation estimator, so 2SDD inherits a precision advantage over estimators that impose a weaker assumption.

Write the untreated potential outcome as

$$Y_{it}(0) \;=\; X_{it}'\gamma + u_{it}, \tag{7}$$

where $X_{it} \in \mathbb{R}^K$ is a nuisance regressor vector (collecting unit and time indicators or their within-transformed analogues) and u_{it} is mean-zero error.

The choice of donor pool is governed by which version of parallel trends the researcher is willing to impose.

A strong version of parallel trends is that model (7) holds for all $(i,t) \in \Omega$, so the full untreated panel $\Omega_0(D)$ is a valid donor pool. This assumption is the baseline condition of BJS2024 and dCDH2024, which is also used in our Section 2. A weaker version of parallel trends is that model (7) holds only for observations in $\Omega_0^*(D) \subset \Omega_0(D)$, where $\Omega_0^*(D)$ consists of never-treated observations and, for each eventually-treated unit, the periods immediately preceding its treatment date. This assumption is implicit in CS2021, who compare changes in outcomes relative to the period immediately prior to treatment for each eventually-treated cohort to those changes for a comparison group.

Let $\tau_1(D) := (Y_{it} - Y_{it}(0))_{(i,t)\in\Omega_1(D)} \in \mathbb{R}^{N_1}$ denote the vector of treatment effects stacked over all treated observations, so that $w'\tau_1(D)$ is their pre-specified weighted average.[18] Under either assumption, the imputation estimator (6) is consistent for $w'\tau_1(D)$, the weighted average treatment effect on the treated, which then converges to its expectation β in probability, provided the donor pool used in the first stage satisfies the relevant restriction.[19] The difference lies in variance: a larger valid donor pool reduces the first-stage estimation error. We can obtain a formal result for the gains in using a larger donor pool under i.i.d. u.

Assumption 4 (Conditional exogeneity and spherical errors)**.** $\mathbb{E}[u \mid X, D] = 0$ *and* $\mathrm{Var}(u \mid X, D) = \sigma^2 I_n$.

[18]The vector $\tau_1(D)$ is also the same as the vector of treatment effects $(\beta_{it})_{(i,t)\in\Omega_1(D)}$ for treated units; we use the "τ" notation here to differentiate from the overall ATT β.

[19]In appendix E, we show that in models without covariates, 2SDD applied to a donor pool consisting of never-treated observations and those for the period just before treatment for treated units is equivalent to the CS2021 and SA2021 estimators.

Proposition 1 (Conditional variance of the imputation estimator)**.** *Suppose Assumption 4 holds and model (7) holds for some donor pool $\tilde{\Omega}_0(D)$. Then,*

$$\mathrm{Var}(\hat{\beta} \mid X, D) \;=\; \underbrace{\sigma^2 w'w}_{\textit{treated-outcome noise}} \;+\; \underbrace{\sigma^2 c'(X_0'X_0)^{-1}c}_{\textit{first-stage estimation error}}\;.$$

The first term is irreducible: it reflects the noise from observing treated outcomes and is unaffected by the choice of donor pool. The second term is the first-stage estimation error and shrinks as more valid untreated donors are added.

Theorem 2 (Larger donor pool weakly lowers variance)**.** *Let $\Omega_R \subseteq \Omega_S$ be two donor pools that both satisfy Assumption 4, with Ω_S arising from a stronger parallel trends restriction. Let $\hat{\beta}_R$ and $\hat{\beta}_S$ denote the imputation estimators using donor pools Ω_R and Ω_S respectively. Then*

$$\mathrm{Var}(\hat{\beta}_S \mid X, D) \;\leq\; \mathrm{Var}(\hat{\beta}_R \mid X, D).$$

3.2 Fixed vs Random Treatments

This subsection explains how our approach to variance estimation gives exact inference when D is random and is conservative while D is fixed. In contrast, the inference procedures in BJS2024 and W2025 resulting from their variance estimation strategies are valid when D is fixed, but can be over-sized when D is random.[20] These observations suggest that one advantage of our approach to inference is that it offers robustness to the possibility that treatment status is random.

Let τ denote a vector of group (or individual) by time treatment effects that are nonstochastic, with the corresponding estimator $\hat{\tau}$. The W2025 approach partitions these treatment effects by group and time, while the BJS2024 approach partitions them at the unit by time level. The ATT is given by $\beta = w'\tau$, which is our target object, where w are weights, which in this case is a function of treatment status D.

There are two possible regimes that we can consider: one where D is fixed and another where D is random (random D does not imply that treatment status is exogenous, only that it is not fixed in repeated samples). We use $\hat{w}$ to denote the estimated weights. When D is fixed, $\hat{w} = w$ because the weights that we plug in are the actual weights for the ATT. When D is random, $\hat{w}$ is an estimator for the target w, so $\hat{w}$ is random while w is not.

[20] BJS2024 develop their inference under the assumption that treatment status is fixed in repeated samples. W2025 notes that it is possible in some software packages to estimate standard errors for average treatment effects that account for uncertainty in treatment assignment. However, the underlying methods used to construct these adjustments are somewhat opaque, and their availability in software packages is platform-specific (for example, the default standard errors produced by the `jwdid` (Rios-Avila, Nagengast and Yotov, 2022) package, a popular implementation of the ETWFE estimator for Stata, take treatment status as fixed).

Based on the derivation of the variance for 2SDD, and the properties of GMM, the variance is asymptotically exact for random D. The crucial assumption is that the second-stage moment condition

$$\mathbb{E}\left[\sum_t (\beta_{it} - \beta) D_{it}\right] = 0$$

must hold for all i for asymptotically exact inference, as the standard inferential theory for GMM relies on mean-zero moments. In general, when D_{it} is nonstochastic, the moment above can be written as

$$\mathbb{E}\left[\sum_t(\beta_{it} - \beta)D_{it}\right] = \mathbb{E}\left[\sum_t(\beta_{it} - \tau_{g(i),t})D_{it}\right] + \sum_t(\tau_{g(i),t} - \beta)D_{it} \neq 0,$$

where $\tau_{g(i),t} = \mathbb{E}(\tau_{it}|G_i = g_i)$, so our asymptotically exact inference result no longer holds. However, it can be shown (see Appendix A) that our variance estimand is conservative:

$$\mathbb{E}\left[\left(\sum_t \left(\left(\beta_{it} - \tau_{g(i),t}\right) + \left(\tau_{g(i),t} - \beta\right)\right) D_{it}\right)^2\right] \geq \mathbb{E}\left[\left(\sum_t \left(\beta_{it} - \tau_{g(i),t}\right) D_{it}\right)^2\right]. \quad (8)$$

By default, BJS2024 and W2025 construct their variance estimators under fixed D. Under the regularity conditions of W2025, $\sqrt{n}\,(\hat{\tau} - \tau)$ is asymptotically jointly normal with non-negligible variance V_τ. Under the fixed D regime, the asymptotic variance of the overall ATT estimator is the variance of

$$\begin{aligned}\sqrt{n}\,(w'\hat{\tau} - w'\tau) &= w'\left(\sqrt{n}\,(\hat{\tau} - \tau)\right) \\ &\xrightarrow{d} N\left(0, w'V_\tau w\right),\end{aligned}$$

The construction of the variance estimator crucially relies on the properties of taking a linear combination of jointly normal random variables, which requires a nonstochastic w. When w is stochastic, this argument is no longer valid. To be precise, with a random D, and $\hat{w} = w + o_P(1)$, by adding and subtracting $\sqrt{n}\hat{w}'\tau$,

$$\begin{aligned}\sqrt{n}\,(\hat{w}'\hat{\tau} - w'\tau) &= \hat{w}'\left(\sqrt{n}\,(\hat{\tau} - \tau)\right) + \left(\sqrt{n}\,(\hat{w} - w)\right)'\tau \\ &= w'\left(\sqrt{n}\,(\hat{\tau} - \tau)\right) + \left(\sqrt{n}\,(\hat{w} - w)\right)'\tau + o_P(1)\end{aligned}$$

so using a variance appropriate only for $\sqrt{n}\,(w'\hat{\tau} - w'\tau)$ does not accurately capture the full variance of the estimator that also contains a non-negligible stochastic component in $\left(\sqrt{n}\,(\hat{w} - w)\right)'\tau$, even asymptotically. If $\hat{w}$ and $\hat{\tau}$ were independent, they would underestimate the true variance. Note that if we were to modify the W2025 variance construction strategy to account for the additional term (through Stata's `margins` command or otherwise), the variance estimator would asymptotically converge to the 2SDD variance estimand.[21]

3.3 Comparing Variance Estimators

In this subsection, we show that the variance estimators for 2SDD, BJS2024 and W2025 can be viewed as variants from a common framework. The analysis in this section is conducted conditional on treatment status and any observed covariates. Since all three estimators share the same point estimate, they all represent the same linear combination of the outcome variable Y_{it}, which follows from using the $\hat{\beta}$ representation of the 2SDD estimator. As a consequence, their variance estimators all take the identical structural form

$$\hat{\sigma}^2 = \sum_{i=1}^{N} \left(\sum_{t:\,(i,t)\in\Omega} w_{it}\, r_{it} \right)^2,$$

where w_{it} are nonstochastic weights, and these variance estimators differ *only* in which residual r_{it} they assign to each observation.

The BJS2024 variance estimator is built around an auxiliary model that assigns a posited treatment effect $\tilde{\tau}_{it}$ to each treated observation, yielding residuals

$$r_{it} = \begin{cases} Y_{it} - \hat{\alpha}_g - \hat{\gamma}_t & (i,t) \in \Omega_0, \\ Y_{it} - \hat{\alpha}_g - \hat{\gamma}_t - \tilde{\tau}_{it} & (i,t) \in \Omega_1, \end{cases}$$

where $\hat{\alpha}$ and $\hat{\gamma}$ are imputed using the sample of untreated observations. The key degree of freedom under this approach is the granularity of the partition used to define $\tilde{\tau}_{it}$: a finer partition shrinks the auxiliary effects towards the cohort-time estimates $\hat{\tau}_{g,t}$, while a coarser partition shrinks them towards the overall ATT $\hat{\beta}$. By default, BJS2024 use $\tilde{\tau}_{it} = \hat{\tau}_{g,t}$, which corresponds to the residual used in the W2025 variance estimator (after applying a finite sample correction). Hence, the 2SDD, BJS2024 and W2025 variance estimators correspond to different choices on this spectrum.

[21]The preceding argument also demonstrates that the variances of the group-time ATTs are the same regardless of whether treatment status is random or fixed in repeated samples. Consequently, the 2SDD approach also offers the flexibility of obtaining exact inference when treatment status is fixed. In this case, an event-study version of 2SDD can be used to obtain group-time-specific ATTs, which can then aggregated to estimate the overall ATT as a linear combination of the estimated coefficients. Then, since the coefficient estimators are jointly normal, the delta method suffices for inference. This approach is identical to W2025, which is exact.

Proposition 2 (Relationship between variance estimators). *(i) The 2SDD GMM variance $\hat{V}$ is numerically equivalent to setting $\tilde{\tau}_{it} = \hat{\beta}$.*

(ii) The W2025 variance estimator is numerically equivalent to setting $\tilde{\tau}_{it} = \hat{\tau}_{g(i),t}$.

Compared to W2025 and the BJS2024 default, even in settings where treatment status is fixed, our approach is more robust to small-sample issues—particularly those arising from having very few observations per cohort. While both variance estimators achieve consistency as the sizes of the treatment cohorts grow without bound, the small-sample performance of the variance estimators can differ regardless of whether treatment status is fixed or random. When the groups are small, the W2025 and BJS2024 default variance estimators, which use $\tilde{\tau}_{it} = \hat{\tau}_{gt}$,can become anti-conservative (which results in size distortions): in the extreme case with each group consisting of only one observation, $\tilde{\tau}_{it} = \hat{\tau}_{it}$, so $r_{it} = 0$. While BJS2024 develop a leave-one-out variation on their variance estimators to address small cohorts, it cannot completely alleviate the problem: with one observation in a group, there are no observations to calculate $\tilde{\tau}_{it}$. In contrast, by avoiding estimating the group-time ATTs, the 2SDD approach provides some protection against these degenerate variances.

Remark 2. An additional advantage of the 2SDD variance estimand is that it is identical to the TWFE variance estimand in the two-period, two-cohort case. Since TWFE is a well-established benchmark with well-understood efficiency properties, 2SDD is a natural extension of this benchmark beyond the 2×2 case. Details are relegated to Appendix B.2.

4 Rejection rates for randomly generated interventions

This section conducts Monte Carlo simulation exercises inspired by Bertrand, Duflo and Mullainathan (2004) to evaluate inference for imputation-style difference-in-differences estimators. The central comparison is among estimators that deliver the same event-study point estimates but differ in how they construct standard errors: the imputation estimator from BJS2024, its leave-out variance version, the W2025 estimator, and the two-stage estimator with and without a finite-sample correction. We use the alternative estimators of CS2021, SA2021, and dCDH2024 as benchmarks for assessing whether the inference procedures in this class are reasonably calibrated in realistic finite samples.

4.1 Data and methodology

Our primary dataset consists of wage data for women between the ages of 25 and 50 from the Current Population Survey (CPS). We define wages as the natural logarithm of weekly earnings, which are recorded in the fourth interview month in the Merged Outgoing Rotation Group of the CPS.[22] The data span a 42-year period from 1979 to 2020 and contain over one million women reporting strictly positive weekly earnings. Using data from 50 states, we construct a state-by-year panel dataset comprising average wages in 2,100 state-year cells.

Our simulation study reports both random-design and fixed-design exercises. The random-design exercise is closest to the theoretical framework in this paper, where treatment paths are part of the stochastic object: we assume that $\{Y_{it}(0), \beta_{it}, D_{it}, X_{it}\}_{t=1}^{T}$ are i.i.d. across individuals. In this exercise, treated states, treatment timing, treatment effects, and outcome innovations are redrawn in each Monte Carlo replication. For comparison, we also report a fixed-design version of the same exercise, in which treated states and treatment timing are drawn once and then held fixed across replications, while outcome innovations and treatment-effect realizations continue to vary. This fixed-design exercise mirrors the repeated-sampling framework emphasized by BJS2024, who primarily consider settings in which treatment status is non-stochastic.

To simulate a staggered treatment setting, we randomly assign states to the treatment group and generate treatments that occur randomly over a specified period. We also consider a non-uniform assignment design based on the treatment assignment probability model from Arkhangelsky et al. (2021). In all cases, we restrict the earliest treatment year to 1982 and the latest treatment year to 2014. Since treatment is an absorbing state, this ensures that we observe outcome data in all treated states for at least 5 years after the treatment event. Standard errors are adjusted for clustering at the state level, following Bertrand, Duflo and Mullainathan (2004).

We estimate the effects of the randomly generated interventions using 2SDD, the same approach with the finite-sample correction, the imputation approach from BJS2024, its leave-out variance version, and the W2025 estimator.[23] These are the primary estimators of interest because they produce numerically identical event-study point estimates in our setting but differ in their approach to inference. We then compare their performance to the alternative estimators of CS2021, SA2021, and dCDH2024.

[22]Using the logarithmic transformation excludes women with zero weekly earnings. While many recent papers use quasi-logarithmic transformations to incorporate zero-valued observations, Thakral and Tô (2023) document substantial biases arising from the use of such transformations, and thus we focus on women with strictly positive earnings following Bertrand, Duflo and Mullainathan (2004).

[23]For the finite-sample corrected version of the GMM-based 2SDD standard errors, we multiply the standard errors by the square root of the standard degrees-of-freedom adjustment used for cluster-robust variance estimators,

$$\left(\frac{N-1}{N-K}\frac{G}{G-1}\right)^{1/2},$$

where N is the number of observations, K is the number of estimated regression degrees of freedom, and G is the number of clusters.

The primary measure we use to evaluate performance is the relative frequency of rejecting the null hypothesis of the true generated effect size at the 5-percent significance level over 1,000 simulations. To summarize size performance across post-treatment periods, we also report the mean absolute deviation from a 5-percent rejection rate. This measure avoids offsetting over-rejection in some periods with under-rejection in others. We also report average standard errors and average per-simulation computational speed. Bias and RMSE are less central for the main comparison because the imputation-style estimators share the same point estimates.

4.2 Simulation results

We conduct an event-study analysis to estimate the effect of the randomly generated interventions in each of the five years starting from the time of treatment. The baseline environment consists of 40 treated states over a 20-year treatment window, with two treated states in each treatment cohort. Treatment effects are heterogeneous and drawn from a normal distribution, with an average value randomly drawn between 2 and 5 percent of the average wage and a standard deviation equal to 10 percent of the average wage. In the simulations reported below, the average true effect is approximately 0.209.

Table 1 reports results from the random-design version of this exercise. We primarily focus on comparing the imputation-style estimators, which deliver the same point estimates but different standard errors. Implementing 2SDD with inference via GMM yields rejection rates close to the 5-percent level in all five post-treatment periods, ranging from 4.6 percent to 6.7 percent, with an average rejection rate of 5.42 percent. Its average standard error is 0.104, and its mean absolute deviation from the nominal rejection rate is only 0.62 percentage points. Applying the finite-sample correction increases the average standard error slightly (from 0.104 to 0.107) and produces rejection rates ranging from 3.5 percent to 6.1 percent, with an average rejection rate of 4.72 percent.

The difference between these results and the default BJS2024 inference procedure is substantial. Although the point estimates are the same, the default imputation standard errors average only 0.076, and the resulting rejection rates range from 15.7 percent to 18.2 percent. Thus, the default imputation variance estimator rejects more than three times as often as the nominal level in this environment. The leave-out version moves in the opposite direction: its standard errors average 0.142 and its rejection rates range from 1.0 percent to 1.7 percent. The W2025 estimator falls between these cases, with average standard errors of 0.088 and rejection rates between 9.0 percent and 12.6 percent.

The benchmark estimators help put these magnitudes in perspective. The CS2021 estimator has rejection rates between 4.3 percent and 5.2 percent, with a mean absolute deviation from the nominal rejection rate of only 0.26 percentage points. The dCDH2024 estimator has rejection rates between 6.2 percent and 7.6 percent, with a mean absolute deviation of 1.88 percentage points. 2SDD and its finite-sample corrected version are therefore much closer to these benchmark methods than to either the default or leave-out imputation variance estimators. The SA2021 estimator is conservative in this setting, with rejection rates between 1.2 percent and 2.4 percent and a mean absolute deviation of 3.02 percentage points.

Table 2 repeats the same exercise under a fixed design, holding treated states and treatment timing fixed across simulations. The qualitative conclusions are unchanged. 2SDD again produces rejection rates close to 5 percent, ranging from 4.4 percent to 5.0 percent, with an average standard error of 0.104. The finite-sample correction is slightly more conservative, with rejection rates ranging from 3.6 percent to 4.3 percent and an average standard error of 0.108.

By contrast, the default imputation variance estimator continues to over-reject sharply even conditional on the realized treatment design. Its rejection rates range from 13.7 percent to 17.2 percent, with an average standard error of 0.076. The leave-out imputation variance estimator remains conservative, with rejection rates between 0.7 percent and 1.8 percent and an average standard error of 0.142. The W2025 estimator again produces standard errors that are too small in this finite-sample setting, with rejection rates between 8.6 percent and 10.1 percent and a mean absolute deviation from the nominal rejection rate of 4.48 percentage points. In comparison, the CS2021 estimator has rejection rates between 3.8 percent and 4.5 percent, with a mean absolute deviation of 0.76 percentage points, while the dCDH2024 estimator has rejection rates between 4.4 percent and 6.5 percent, with a mean absolute deviation of 1.02 percentage points. The SA2021 estimator remains conservative, with rejection rates between 1.1 percent and 1.7 percent and a mean absolute deviation of 3.52 percentage points.

Finally, Table 3 reports results using the non-uniform assignment mechanism from Arkhangelsky et al. (2021). This exercise is designed to create a stronger relationship between treatment assignment and state-level economic characteristics. The GMM-based variance estimator continues to perform well. The uncorrected version has rejection rates between 4.9 percent and 6.4 percent, with an average rejection rate of 5.68 percent and a mean absolute deviation from the nominal rejection rate of 0.72 percentage points. The finite-sample corrected version has rejection rates between 3.8 percent and 5.3 percent, with an average rejection rate of 4.76 percent and a mean absolute deviation of 0.44 percentage points. The corresponding average standard errors are both approximately 0.019.

The same exercise again reveals large differences across inference procedures. The default imputation variance estimator has an average standard error of only 0.013 and rejection rates between 16.0 percent and 18.4 percent, with an average rejection rate of 17.04 percent and a mean absolute deviation from the nominal rejection rate of 12.04 percentage points. The leave-out version has an average standard error of 0.026 and rejection rates between 1.1 percent and 1.6 percent, with an average rejection rate of 1.42 percent and a mean absolute deviation of 3.58 percentage points. The W2025 estimator has rejection rates between 9.8 percent and 12.1 percent, with an average rejection rate of 10.86 percent and a mean absolute deviation of 5.86 percentage points. The benchmark estimators again place the GMM-based variance approach in a favorable light: CS2021 has rejection rates between 3.9 percent and 5.0 percent, with an average rejection rate of 4.48 percent and a mean absolute deviation of 0.52 percentage points; dCDH2024 has rejection rates between 4.3 percent and 6.8 percent, with an average rejection rate of 5.74 percent and a mean absolute deviation of 1.02 percentage points; and SA2021 is again conservative, with rejection rates between 1.5 percent and 2.5 percent, an average rejection rate of 1.90 percent, and a mean absolute deviation of 3.10 percentage points.

Across these different simulation environments with finite samples, we find that the default BJS2024 variance estimator is too small, while the leave-out version is too conservative. The GMM-based variance estimator, both with and without the finite-sample correction, delivers rejection rates close to the benchmark methods, while retaining the computational simplicity of the imputation-style approach and the efficiency advantage that BJS2024 establish for the underlying point estimator.

5 Empirical applications

This section applies the two-stage estimator to published empirical settings and compares the resulting event-study estimates with those from alternative difference-in-differences methods. We use the school finance reform application from Lafortune, Rothstein and Schanzenbach (2018) as the main empirical illustration because it is closely connected to our simulation design. We then summarize the results from a broader set of replication exercises.

5.1 School finance reforms

Lafortune, Rothstein and Schanzenbach (2018) study the effect of court-ordered and legislative school finance reforms on the distribution of school resources across districts. A central motivation for these reforms was to generate "higher spending in low-income than in high-income districts, to compensate for the out-of-school disadvantages that low-income students face." This setting is therefore especially useful for comparing estimators because the economic interpretation depends not only on whether school finance reforms increased revenues on average, but also on whether the increases were concentrated among lower-income districts.

We emphasize this application in the main text due to its close connection to our simulation design. The school finance reform example is also highlighted in the recent event-study guide of Miller (2023), and motivates our baseline simulation environment with treatment staggered over 20 years. The dataset contains a relatively large number of small treatment cohorts, with 11 cohorts consisting of a single state, 6 cohorts consisting of two states, and the remaining cohort consisting of three states, making the application a useful empirical counterpart to the finite-sample issues studied in Section 4.

The original paper documents that school finance reforms led to large increases in state revenues and total revenues for low-income districts, while the estimated effects for high-income districts are smaller. This pattern is central for the interpretations in the paper, which suggests that school finance reforms increased the progressivity of school funding because they largely shifted resources toward districts serving lower-income students rather than simply increasing state aid uniformly.

We compare 2SDD with the default BJS2024 imputation variance estimator, the W2025 estimator, and the alternative heterogeneity-robust estimators from CS2021, SA2021, and dCDH2024.[24] We also report the corresponding TWFE estimates as a point of comparison.

Most methods agree that reforms produced a sustained increase in state transfers per pupil in the lowest-income districts, consistent with the central finding of Lafortune, Rothstein and Schanzenbach (2018). Thus, we focus on their result on the effect on state revenues in the highest-income districts.[25]

Using 2SDD, none of the post-reform estimates are statistically significant at the 5 percent level (Figure 1). The same conclusion holds for the CS2021, SA2021, dCDH2024, and TWFE estimators. In contrast, the default BJS2024 variance estimator indicates a significant increase in state revenues for five of the first nine years following the reform, while the W2025 estimator indicates a significant increase in three of those years, despite having the same point estimates as 2SDD.

The BJS2024 and W2025 standard errors make the highest-income districts appear to experience statistically significant gains in state aid in several of the early post-reform periods. Such a conclusion would weaken the interpretation that the reforms primarily operated by directing additional resources toward low-income districts. By contrast, 2SDD

[24] In this application, the leave-out variance estimator from BJS2024 is not feasible because many treatment cohorts contain only a single state.

[25] Figure A2 reports the full set of estimates applying each of the estimators to all of the event-study analyses in Lafortune, Rothstein and Schanzenbach (2018).

and the benchmark heterogeneity-robust estimators preserve the central message of Lafortune, Rothstein and Schanzenbach (2018): reforms increased state support for school districts, but the evidence for meaningful increases in state aid is concentrated among lower-income districts.

5.2 Additional empirical applications

We also conduct a broader replication exercise for all of the empirical papers with variation in treatment timing and a single treatment event listed in Table 1 of SA2021. We reproduce the first event-study specification for each main empirical setting in Figure A1 and report the remaining event-study figures in Figures A2 to A7. These figures show that, except for TWFE, the different heterogeneity-robust methods generally produce fairly similar point estimates. The main differences arise in the associated confidence intervals and the flexibility of implementing the 2SDD approach. Across these applications, 2SDD preserves the transparency and computational simplicity of regression-based DD while avoiding the more volatile inference patterns that some alternative heterogeneity-robust methods produce.

Standard errors. We find that 2SDD and CS2021 produce comparable standard errors in four of the settings, but CS2021 yields substantially larger standard errors in the two large county-level applications (Figure A8 and tables A2 and A3). The SA2021 and dCDH2024 estimators can be comparatively precise when the number of groups is large, but they produce much larger standard errors in applications with many small treatment cohorts. The BJS2024 estimator produces smaller standard errors than 2SDD in five of the six settings summarized in Table A2, while the Bailey and Goodman-Bacon (2015) application illustrates how the reverse can also occur.

When comparing outliers across the different estimators and approaches to inference, 2SDD appears to provide a practical middle ground (Supp. Appendix A.4). 2SDD generates the fewest outlier standard errors, and the outliers that do appear generally reflect conservative rather than overly precise standard errors (Figure A9). The outliers for BJS2024, SA2021, and dCDH2024 mostly arise because those methods are unusually precise in particular applications. By contrast, CS2021 generates the most outliers, nearly all of which reflect imprecision. The corresponding t-statistic comparisons (Figure A10 and tables A5 to A7) show that BJS2024, SA2021, and dCDH2024 produce larger absolute t-statistics and a higher share of statistically significant event-study coefficients than 2SDD. In addition, 2SDD has the lowest frequency of instances in which the difference between its t-statistic and the average of the other methods' t-statistics falls in the tails of the distribution.

Flexibility of implementation. One of the replication exercises illustrates a practical advantage of the two-stage framework that is not captured by comparisons of standard errors alone. In Kuziemko, Meckel and Rossin-Slater (2018), treatment timing is observed by month, but the event-study estimand is expressed in years since treatment, with years defined as 12-month intervals relative to the exact month of transition. The 2SDD implementation accommodates this structure directly: the first stage can include month fixed effects, while the second stage includes the coarser year-since-treatment indicators that define the estimands of interest. In contrast, the off-the-shelf implementations of several alternative heterogeneity-robust estimators require either coarsening treatment timing to the treatment year or estimating finer monthly treatment effects and then aggregating them into annual bins. The latter approach is conceptually possible, but it introduces additional implementation and inference complications. This example highlights a broader benefit of 2SDD: because the nuisance adjustment and treatment-effect aggregation are separated across stages, researchers can tailor the second-stage estimands to the empirical question while retaining a rich first-stage specification for untreated potential outcomes.

Pre-trends and TWFE comparisons. We also record two broader lessons that are useful for applied work. First, the pre-event coefficients should be interpreted carefully. The BJS2024 first-stage approach and the second-stage approach used in the main 2SDD event-study figures estimate different pre-event objects, even though both can be used as tests of parallel trends under the null. These different choices matter empirically, as the first-stage approach produces more outliers among pre-event standard errors (Figure A11 and table A8). Second, the comparison with TWFE (Supp. Appendix A.6) suggests that researchers should not default to TWFE in staggered-adoption event studies. For nearly one-sixth of the dynamic treatment effect estimates in the replicated applications, the conclusion based on TWFE does not align with the conclusion based on any of the heterogeneity-robust estimators. In 29 percent of these discrepancies, all of the heterogeneity-robust estimators are statistically significant with the same sign while the TWFE estimate is not significantly different from zero. In 58 percent of these discrepancies, TWFE is statistically significant while all of the heterogeneity-robust estimators are not significantly different from zero.

6 Conclusion

When adoption of a treatment is staggered across time, and the average effects of the treatment vary by group and period, the usual difference-in-differences regression specification does not identify an easily interpretable measure of the typical effect of the treatment. When the duration-specific effects are also heterogeneous, neither do the coefficients from the usual event-study specification. The ultimate source of these identification failures is that outcomes are not necessarily linear in group, period, and treatment status, as difference-in-differences and event-study regression specifications assume.

The two-stage approach developed in this paper is motivated by the observation that, under parallel trends, untreated outcomes are linear in group and period effects. Those effects are therefore identified from a first-stage regression estimated using the sample of untreated observations. The average effect of the treatment on the treated is then identified from a regression of outcomes on treatment status, after removing group and period effects. This procedure transparently handles the complexities of staggered treatment adoption with familiar and straightforward tools, analogous to traditional regression methods. Estimation and inference are simple and intuitive, and can be easily extended to a variety of different treatment effect measures, including event studies, group-specific treatment effects, design-based analyses, continuous treatments, and triple-difference analyses. The empirical applications also illustrate this flexibility in practice: for example, when treatment timing is observed at a finer frequency than the desired event-study estimands, 2SDD can use the finer time scale in the first stage while estimating coarser event-time effects in the second stage.

Monte Carlo simulations demonstrate that the two-stage estimator correctly identifies informative average treatment effect measures, and the associated approach to inference has size closer to the nominal rate than the more complex and computationally demanding alternative methods. The empirical application to school finance reforms (Lafortune, Rothstein and Schanzenbach, 2018) provides a clear example in which different variance estimators can change the substantive interpretation of an event-study design. Across a broader range of replication exercises, 2SDD produces comparatively stable standard errors and t-statistics, while some alternative methods generate more frequent outliers or conclusions that are more sensitive to the choice of inference procedure, supporting our approach to estimation and inference as a viable and effective option for applied research. Our approach also facilitates adaptations to a variety of problems, and indeed, the general approach proposed in this paper has already been developed by other authors to address settings where time-varying covariates are affected by the treatment (Caetano et al., 2022), to interactive fixed effects models (Brown and Butts, 2023), and to local-projections estimation (Dube et al., 2023).

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Table 1: Simulations (CPS wage data, heterogeneous treatment effects, random design): 40 states treated over 20 years (2 per year)

	Rejection rate by event time							
Method	0	1	2	3	4	Size dist.	S.E.	Speed
2SDD	6.7	4.6	5.3	4.9	5.6	0.62	0.104	0.14
2SDD (f.s.c.)	6.1	3.5	5.0	4.4	4.6	0.72	0.107	0.26
BJS	18.2	16.4	16.8	15.7	16.3	11.68	0.076	0.21
BJS (leave out)	1.7	1.0	1.6	1.5	1.2	3.60	0.142	0.21
W	12.6	9.0	11.0	9.0	10.9	5.50	0.088	19.68
CS	4.8	4.3	4.9	5.2	5.1	0.26	0.143	49.64
dCDH (dyn)	6.9	6.2	7.5	7.6	6.2	1.88	0.138	11.82
SA	2.0	1.2	2.3	2.4	2.0	3.02	0.166	142.21

Note: The table reports results from 1,000 simulations of 40 treated states over 20 years, with two treated states in each treatment year. The data consist of log wages for women between the ages of 25 and 50 from the CPS. Treatment effects are heterogeneous and drawn from a normal distribution, with an average value drawn uniformly at random between 2 and 5 percent of the average wage and a standard deviation equal to 10 percent of the average wage. Rejection rates are reported in percentage points. Size dist. is the mean absolute deviation of the five post-treatment rejection rates from the nominal 5 percent level. S.E. denotes the standard error averaged across post-treatment periods. Speed denotes average per-simulation runtime in seconds using the corresponding Stata package. 2SDD refers to the two-stage method proposed in the current paper; 2SDD (f.s.c.) applies the finite-sample correction. BJS and BJS (leave out) refer to the default and leave-out variance estimators from Borusyak, Jaravel and Spiess (2024). W, CS, SA, and dCDH (dyn) refer to Wooldridge (2025), Callaway and Sant'Anna (2021), Sun and Abraham (2021), and de Chaisemartin and d'Haultfoeuille (2024), respectively.

Table 2: Simulations (CPS wage data, heterogeneous treatment effects, fixed design): 40 states treated over 20 years (2 per year)

	Rejection rate by event time							
Method	0	1	2	3	4	Size dist.	S.E.	Speed
2SDD	4.4	4.9	5.0	4.5	4.6	0.32	0.104	0.08
2SDD (f.s.c.)	3.6	3.9	4.3	4.0	4.1	1.02	0.108	0.25
BJS	15.0	17.2	14.8	13.7	14.5	10.04	0.076	0.20
BJS (leave out)	1.0	0.9	1.8	0.7	1.7	3.78	0.142	0.21
W	10.1	10.1	9.4	8.6	9.2	4.48	0.089	20.59
CS	4.2	4.4	3.8	4.5	4.3	0.76	0.144	48.85
dCDH (dyn)	6.1	5.6	6.3	4.4	6.5	1.02	0.138	9.51
SA	1.5	1.6	1.7	1.5	1.1	3.52	0.168	150.78

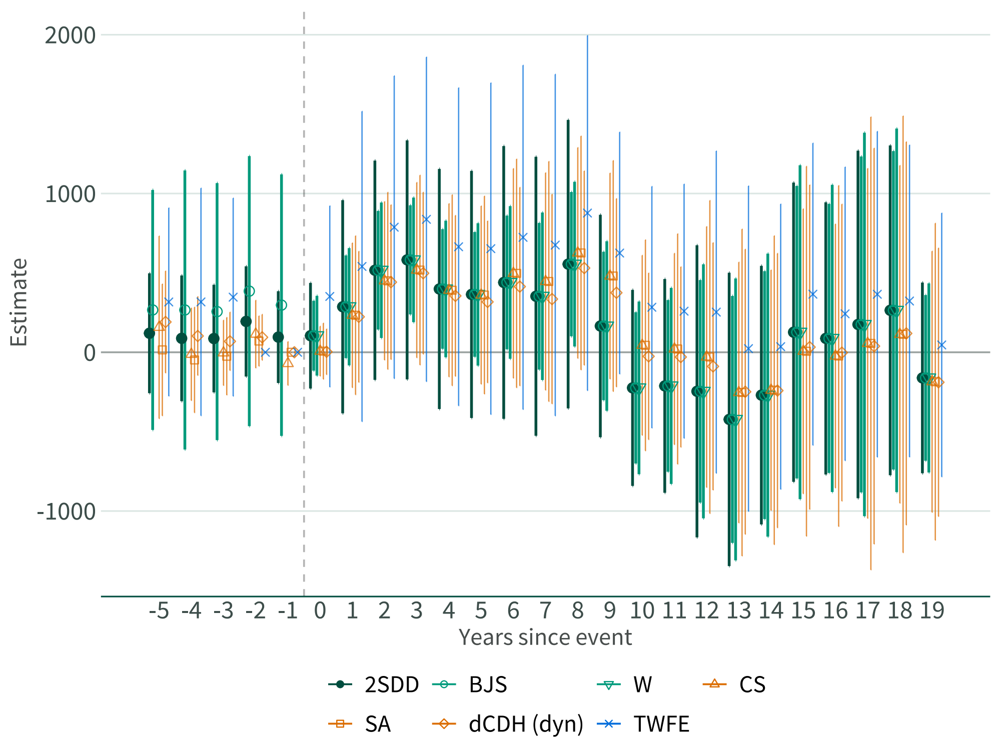
Note: The table reports the fixed-design analog of Table 1. Treatment states and treatment timing are drawn once and then held fixed across the 1,000 simulations. See the note accompanying Table 1 for further information.

Table 3: Simulations (CPS wage data, heterogeneous treatment effects, non-uniform assignment): 40 states treated over 20 years (2 per year)

	Rejection rate by event time							
Method	0	1	2	3	4	Size dist.	S.E.	Speed
2SDD	5.9	5.8	6.4	5.4	4.9	0.72	0.019	0.27
2SDD (f.s.c.)	5.2	4.8	4.7	5.3	3.8	0.44	0.019	0.76
BJS	18.4	16.5	17.2	17.1	16.0	12.04	0.013	0.39
BJS (leave out)	1.4	1.5	1.1	1.6	1.5	3.58	0.026	0.64
W	10.8	12.1	9.8	10.2	11.4	5.86	0.016	72.33
CS	4.5	3.9	5.0	4.7	4.3	0.52	0.025	49.97
dCDH (dyn)	6.3	5.5	6.8	4.3	5.8	1.02	0.024	14.10
SA	2.5	2.2	1.7	1.6	1.5	3.10	0.029	118.38

Note: The table reports results from 1,000 simulations using the non-uniform treatment assignment probability model from Arkhangelsky et al. (2021). This assignment mechanism allows treatment timing to be correlated with state-level characteristics. See the note accompanying Table 1 for further information.

Figure 1: Empirical application: State revenues in high-income districts



Note: This figure reports event-study estimates from applying each estimator to Figure 4 of Lafortune, Rothstein and Schanzenbach, 2018. The outcome is state revenues per pupil in the highest-income districts. The figure compares 2SDD with the default imputation variance estimator from BJS2024, the estimator from W2025, the estimators from CS2021, SA2021, and dCDH2024, and TWFE. Confidence intervals are reported at the 95 percent level.

A Proofs

Proof of Lemma 1.

Since $\beta_{it} := Y_{it} - Y_{it}(0)$ and D_{it} is binary, under our assumptions, $Y_{it} = Y_{it}(0) + \beta_{it} D_{it}$. Therefore, letting $\bar{U}_i^0 := \frac{1}{T_i^0}\sum_{t=1}^{T_i^0}\left(Y_{it}(0) - X_{it}'\gamma\right)$,

$$
\begin{aligned}
\mathbb{E}\left[\sum_t D_{it}\tilde{\varepsilon}_{it}\right] &= \mathbb{E}\left[\sum_t D_{it}\left(\varepsilon_{it} - \frac{1}{T_i^0}\sum_{t=1}^{T_i^0}\varepsilon_{it}\right)\right] \\
&= \mathbb{E}\left[\sum_t D_{it}\left(Y_{it} - D_{it}\beta - X_{it}'\gamma - \frac{1}{T_i^0}\sum_{t=1}^{T_i^0}\left(Y_{it} - X_{it}'\gamma\right)\right)\right] \\
&= \mathbb{E}\left[\sum_t D_{it}\left(Y_{it} - X_{it}'\gamma - \bar{U}_i^0\right) - \sum_t D_{it}\beta\right] \\
&= \mathbb{E}\left[\sum_t D_{it}\mathbb{E}\left[\left(Y_{it}(0) + \beta_{it}D_{it} - X_{it}'\gamma - \bar{U}_i^0\right) \mid \{D_{it}\}_{t=1}^T\right]\right] - \sum_t \mathbb{E}\left[D_{it}\beta\right] \\
&\overset{(A1)}{=} \mathbb{E}\left[\sum_t D_{it}\mathbb{E}\left[\left(\lambda_i + \beta_{it}D_{it} - \lambda_i\right) \mid \{D_{it}\}_{t=1}^T\right]\right] - \sum_t \mathbb{E}\left[D_{it}\beta\right] \\
&= \mathbb{E}\left[\sum_t D_{it}\left(\beta_{it}D_{it}\right) - \sum_t D_{it}\beta\right] = \mathbb{E}\left[\sum_t \beta_{it}D_{it}\right] - \sum_t \mathbb{E}\left[D_{it}\right]\frac{\sum_t \mathbb{E}\left[\beta_{it}D_{it}\right]}{\sum_t \mathbb{E}\left[D_{it}\right]} = 0.
\end{aligned}
$$

Similarly, letting $\bar{X}_i^0 := \frac{1}{T_i^0}\sum_{t=1}^{T_i^0} X_{it}$,

$$
\begin{aligned}
&\mathbb{E}\left[\sum_t \tilde{X}_{it}\tilde{\varepsilon}_{it} \mid D_{it} = 0\right] = \mathbb{E}\left[\sum_t \tilde{X}_{it}\left(\varepsilon_{it} - \frac{1}{T_i^0}\sum_{t=1}^{T_i^0}\varepsilon_{it}\right) \mid D_{it} = 0\right] \\
&= \mathbb{E}\left[\sum_{t=1}^{T_i^0}\left(X_{it} - \bar{X}_i^0\right)\left(Y_{it} - X_{it}'\gamma - \frac{1}{T_i^0}\sum_{t=1}^{T_i^0}\left(Y_{it} - X_{it}'\gamma\right)\right) \mid D_{it} = 0\right] \\
&= \mathbb{E}\left[\sum_{t=1}^{T_i^0}\left(X_{it}\left(Y_{it}(0) - X_{it}'\gamma\right) - \bar{X}_i^0\left(Y_{it}(0) - X_{it}'\gamma\right)\right) \mid D_{it} = 0\right] \\
&\quad - \mathbb{E}\left[\sum_{t=1}^{T_i^0} X_{it}\bar{U}_i^0 - \sum_{t=1}^{T_i^0}\bar{X}_i^0\bar{U}_i^0 \mid D_{it} = 0\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\sum_{t=1}^{T_i^0}\left(X_{it}\left(Y_{it}(0) - X_{it}'\gamma\right) - \bar{X}_i^0\left(Y_{it}(0) - X_{it}'\gamma\right)\right) \mid \{X_{it}\}_{t=1}^T, \{D_{it}\}_{t=1}^T\right] \mid D_{it} = 0\right] \\
&\quad - \mathbb{E}\left[\mathbb{E}\left[\sum_{t=1}^{T_i^0} X_{it}\bar{U}_i^0 - \sum_{t=1}^{T_i^0}\bar{X}_i^0\bar{U}_i^0 \mid \{X_{it}\}_{t=1}^T, \{D_{it}\}_{t=1}^T\right] \mid D_{it} = 0\right]
\end{aligned}
$$

$$\overset{(A1)}{=} \mathbb{E}\left[\sum_{t=1}^{T_i^0}\left(X_{it}\lambda_i - \bar{X}_i^0\left(\lambda_i\right)\right) - \sum_{t=1}^{T_i^0} X_{it}\left(\frac{1}{T_i^0}\sum_{t=1}^{T_i^0}\left(\lambda_i\right)\right) + \sum_{t=1}^{T_i^0}\bar{X}_i^0\left(\frac{1}{T_i^0}\sum_{t=1}^{T_i^0}\left(\lambda_i\right)\right) \mid D_{it}=0\right]$$

$$= \mathbb{E}\left[\sum_{t=1}^{T_i^0}\left(X_{it}\lambda_i - \bar{X}_i^0\lambda_i\right) - \sum_{t=1}^{T_i^0} X_{it}\lambda_i + \sum_{t=1}^{T_i^0}\bar{X}_i^0\lambda_i \mid D_{it}=0\right] = 0.$$

Lemma A.1. *Under Assumptions 1-3, $\hat{\gamma} \xrightarrow{p} \gamma$ and $\hat{\beta} \xrightarrow{p} \beta$.*

Proof of Lemma A.1. We have

$$\tilde{Y}_{0i} - \tilde{X}_{0i}\gamma = \begin{bmatrix} \left(\tilde{\varepsilon}_{i1}\right)\left(1-D_{i1}\right) \\ \vdots \\ \left(\tilde{\varepsilon}_{iT}\right)\left(1-D_{iT}\right) \end{bmatrix} =: \tilde{\varepsilon}_{0i}.$$

Hence,

$$\hat{\gamma} = \gamma + \left(\frac{1}{N}\sum_i \tilde{X}_{0i}'\tilde{X}_{0i}\right)^{-1}\left(\frac{1}{N}\sum_i \tilde{X}_{0i}'\tilde{\varepsilon}_{0i}\right).$$

The rank conditions in Assumption 2 ensure that the limit of the denominator is invertible. Lemma 1 implies that the first moment of the numerator is indeed zero due to Assumption 1. To apply the weak law of large numbers (WLLN) for i.i.d. observations and the continuous mapping theorem (CMT) on the respective objects, we just need bounded second moments. In particular, with $\varepsilon_{it} = Y_{it}(0)$, $\mathbb{E}[\varepsilon_{it}^4] \le C$, $\mathbb{E}[X_{kit}^8] \le C$ from Assumption 3 imply $\mathbb{E}[\tilde{\varepsilon}_{it}^4] < \infty$ and $\mathbb{E}\left\|\tilde{X}_{0i}'\tilde{X}_{0i}\right\|^4 < \infty$ by repeated applications of the Cauchy-Schwarz inequality. Hence, $\mathbb{E}\left\|\tilde{X}_{0i}'\tilde{X}_{0i}\right\|^2 \le C$ and $\mathbb{E}\left\|\tilde{X}_{0i}'\tilde{\varepsilon}_{0i}\right\|^2 \le \mathbb{E}\left\|\tilde{X}_{0i}'\tilde{X}_{0i}\right\|\mathbb{E}[\tilde{\varepsilon}_{0i}^2] \le C$, so WLLN and CMT imply that $\hat{\gamma} \xrightarrow{p} \gamma$.

The OLS estimator is:

$$\begin{aligned}
\hat{\beta} &= \left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\right)^{-1}\left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\left(\tilde{Y}_{it} - \tilde{X}_{it}\hat{\gamma}\right)\right) \\
&= \left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\right)^{-1}\left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\left(\beta_{it}D_{it} + \tilde{\varepsilon}_{it} - \tilde{X}_{it}'\left(\hat{\gamma}-\gamma\right)\right)\right) \\
&= \left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\right)^{-1}\left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\beta_{it}\right) + \left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\right)^{-1}\left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\left(\tilde{\varepsilon}_{it} - \tilde{X}_{it}'\left(\hat{\gamma}-\gamma\right)\right)\right).
\end{aligned}$$

Using WLLN, $\frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it} \xrightarrow{p} \mathbb{E}[\sum_t D_{it}]$ and $\frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\beta_{it} \xrightarrow{p} \mathbb{E}[\sum_t D_{it}\beta_{it}]$, so $\left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\right)^{-1}\left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\beta_{it}\right) \xrightarrow{p} \beta$. Applying a similar argument for the second term, since both $D_{it}\tilde{\varepsilon}_{it}$ and $D_{it}\tilde{X}_{it}$ have bounded moments $\left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\right)^{-1}\left(\sum_{i=1}^{N}\sum_{t=1}^{T} D_{it}\tilde{\varepsilon}_{it}\right) \xrightarrow{p}$

0 and $\left(\sum_{i=1}^N \sum_{t=1}^T D_{it}\right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T D_{it} \left(\tilde{X}_{it}' \left(\hat{\gamma}-\gamma\right)\right)\right) = O_P(1) o_P(1) = o_P(1)$. Hence, $\hat{\beta} \xrightarrow{p} \beta$.

□

Proof of Theorem 1. If the conditions of Theorem 6.1 of Newey and McFadden (1994) are satisfied, the result automatically follows. Hence, the proof verifies its conditions. Due to Lemma A.1, we already have $\hat{\gamma} \xrightarrow{p} \gamma$ and $\hat{\beta} \xrightarrow{p} \beta$, fulfilling the probability limit requirement. Next, we want to show the following:

1. β is in the interior of the parameter space.
2. $g(Z; \gamma, \beta)$ is continuously differentiable around β.
3. $\mathbb{E}[g(Z; \gamma, \beta)] = 0$ and $\mathbb{E}\left[\|g(Z; \gamma, \beta)\|^2\right]$ is finite.
4. $\mathbb{E}\left[\sup_{(\gamma,\beta)} \|\nabla g(Z; \gamma, \beta)\|\right] < \infty$, where $\nabla g(Z; \gamma, \beta)$ is the derivative of g with respect to (γ', β).
5. $\mathbb{E}[\nabla g(Z; \gamma, \beta)]' \mathbb{E}[\nabla g(Z; \gamma, \beta)]$ is nonsingular.
6. $\frac{1}{N} \sum_{i=1}^N g\left(Z_i; \hat{\gamma}, \beta\right) \xrightarrow{p} 0$ and $\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \left(\tilde{X}_{0i}' \left(\tilde{Y}_{0i} - \tilde{X}_{0i}\gamma\right)\right) \xrightarrow{p} 0$.

Condition 1 is straightforward as long as no further constraints are imposed on β, which is true in this setting. For condition 2, observe that $\nabla_\beta g(Z; \gamma, \beta) = -\sum_t D_t$, which is continuously differentiable. For condition 3, $\mathbb{E}[g(Z; \gamma, \beta)] = 0$ is immediate by assumption, and we have

$$\begin{aligned}
\mathbb{E}\left[\|g(Z; \gamma, \beta)\|^2\right] &= \mathbb{E}\left[\left(\sum_{t=1}^T D_{it}\left(\tilde{Y}_{it} - \tilde{X}_{it}'\gamma - D_{it}\beta\right)\right)^2\right] \\
&= \mathbb{E}\left[\left(\sum_{t=1}^T \left[\tilde{\varepsilon}_{it} + \left(\beta_{it} - \beta\right) D_{it}\right] D_{it}\right)^2\right] < \infty
\end{aligned}$$

due to those objects' finite moments and T being finite. Condition 4 is immediate from finite moments, and condition 5 is immediate from the rank condition. For condition 6,

$$\begin{aligned}
\frac{1}{N}\sum_{i=1}^N g(Z_i; \hat{\gamma}, \beta) &= \frac{1}{N}\sum_{i=1}^N \sum_{t=1}^T D_{it}\left(\tilde{Y}_{it} - \tilde{X}_{it}\hat{\gamma} - D_{it}\beta\right) \\
&= \frac{1}{N}\sum_{i=1}^N \sum_{t=1}^T \left[\beta_{it} D_{it} + \tilde{\varepsilon}_{it} - \tilde{X}_{it}'\left(\hat{\gamma} - \gamma\right) - D_{it}\beta\right] D_{it} \\
&= \frac{1}{N}\sum_{i=1}^N \sum_{t=1}^T \left[\tilde{\varepsilon}_{it} - \tilde{X}_{it}'\left(\hat{\gamma} - \gamma\right) + \left(\beta_{it} - \beta\right) D_{it}\right] D_{it} = o_P(1)
\end{aligned}$$

due to previous arguments. Finally, the second part of condition 6 is immediate from the WLLN. Hence, we obtain the normality result and it remains to show that the variance estimator is consistent.

By CMT, we can use γ, β in place of $\hat{\gamma}, \hat{\beta}$ in the variance since moments of random variables are bounded

$$\hat{V} = G_{\beta}^{-1} \left(\frac{1}{N} \sum_i \mathbb{E}\left[\left(g_i + G_\gamma \psi_i\right) \left(g_i + G_\gamma \psi_i\right)' \right] \right) G_{\beta}^{-1} + o_P(1).$$

It suffices to consider the meat, since $\hat{G}_\beta$ converges to G_β:

$$g_i + G_\gamma \psi_i = \sum_t D_{it} \left(\tilde{Y}_{it} - \tilde{X}_{it}' \gamma - \beta D_{it} \right) + G_\gamma \mathbb{E}\left[\tilde{X}_{0i}' \tilde{X}_{0i} \right]^{-1} \tilde{X}_{0i}' \left(\tilde{Y}_{0i} - \tilde{X}_{0i} \gamma \right).$$

Since observations are independent across i, applying WLLN results in the consistency result. To apply the WLLN, the sufficient conditions are $\mathbb{E}\left\| \tilde{X}_{0i}' \tilde{\varepsilon}_{0i} \right\|^4 < \infty$, $\mathbb{E}\tilde{\varepsilon}_{it}^4 < \infty$, $\mathbb{E}(\beta_{it} - \beta)^4 < \infty$, and $\mathbb{E}\left\| \tilde{X}_{0i}' \tilde{X}_{0i} \right\|^4 < \infty$. Note that $\mathbb{E}\tilde{\varepsilon}_{it}^4 < \infty$, and $\mathbb{E}(\beta_{it} - \beta)^4 < \infty$ result from how $\tilde{Y}_{it}$ is also squared. Since $\mathbb{E}[\beta_{it}^4] \leq C$ implies $\mathbb{E}(\beta_{it} - \beta)^4 < \infty$, $\mathbb{E}[X_{kit}^8] \leq C$ implies $\mathbb{E}\left\| \tilde{X}_{0i}' \tilde{X}_{0i} \right\|^4 < \infty$, and $\mathbb{E}\left\| \tilde{X}_{0i}' \tilde{\varepsilon}_{0i} \right\|^4 \leq \mathbb{E}\left\| \tilde{X}_{0i}' \tilde{X}_{0i} \right\|^4 \mathbb{E}[\tilde{\varepsilon}_{it}^4]$ by Cauchy-Schwarz inequality, the sufficient conditions are satisfied and the variance estimator is consistent. □

Proof of Proposition 1. Define the stacked error vectors $u_1 := (u_{it})_{(i,t) \in \tilde{\Omega}_1(D)} \in \mathbb{R}^{N_1}$ and $u_0 := (u_{it})_{(i,t) \in \tilde{\Omega}_0(D)} \in \mathbb{R}^{N_0}$, so that model (2) implies $Y_0 = X_0 \gamma + u_0$ and $Y_1 = X_1 \gamma + u_1 + \tau_1(D)$.

Substituting these into the imputation estimator (3) and subtracting $w' \tau_1(D)$ gives

$$\begin{aligned}
\hat{\theta} - w' \tau_1(D) &= w'(Y_1 - X_1 \hat{\gamma}) - w' \tau_1(D) \\
&= w'(X_1 \gamma + u_1 + \tau_1(D) - X_1 \hat{\gamma}) - w' \tau_1(D) \\
&= w' u_1 - w' X_1 (\hat{\gamma} - \gamma) \\
&= w' u_1 - c'(\hat{\gamma} - \gamma),
\end{aligned}$$

where $c := X_1' w \in \mathbb{R}^K$.

Since $\hat{\gamma} = (X_0' X_0)^{-1} X_0' Y_0 = \gamma + (X_0' X_0)^{-1} X_0' u_0$ by model (2), we have $\hat{\gamma} - \gamma = (X_0' X_0)^{-1} X_0' u_0$, so

$$\hat{\theta} - w' \tau_1(D) = w' u_1 - c'(X_0' X_0)^{-1} X_0' u_0.$$

We now compute the conditional variance. Under the assumption spherical variances,

$\text{Var}(u \mid X, D) = \sigma^2 I_n$, which implies $u_1 \perp u_0 \mid X, D$. We obtain

$$\begin{aligned}
\text{Var}(\hat{\theta} \mid X, D) &= \text{Var}(w'u_1 \mid X, D) + \text{Var}(c'(X_0'X_0)^{-1}X_0'u_0 \mid X, D) \\
&= \sigma^2 w'w + \sigma^2 c'(X_0'X_0)^{-1}X_0'X_0(X_0'X_0)^{-1}c \\
&= \sigma^2 w'w + \sigma^2 c'(X_0'X_0)^{-1}c,
\end{aligned}$$

where the second equality uses $\text{Var}(w'u_1 \mid X, D) = \sigma^2 w'w$ and $\text{Var}(X_0'u_0 \mid X, D) = \sigma^2 X_0'X_0$, and the third simplifies $(X_0'X_0)^{-1}X_0'X_0(X_0'X_0)^{-1} = (X_0'X_0)^{-1}$. □

Proof of Theorem 2. By Proposition 1, the variances are $\sigma^2 w'w + \sigma^2 c'(Z_S'Z_S)^{-1}c$ and $\sigma^2 w'w + \sigma^2 c'(Z_R'Z_R)^{-1}c$ respectively. Since $\Omega_R \subseteq \Omega_S$, writing $\Omega_A = \Omega_S \setminus \Omega_R$ gives $Z_S'Z_S = Z_R'Z_R + Z_A'Z_A \succeq Z_R'Z_R$ in the Löwner order. Matrix inversion reverses the Löwner order on positive-definite matrices, so $(Z_S'Z_S)^{-1} \preceq (Z_R'Z_R)^{-1}$, and therefore $c'(Z_S'Z_S)^{-1}c \leq c'(Z_R'Z_R)^{-1}c$. □

Proof of Equation (8). Our variance estimand is:

$$\begin{aligned}
\mathbb{E}\left[\left(\sum_t \left((\beta_{it} - \tau_{gp}) + (\tau_{gp} - \beta)\right) D_{it}\right)^2\right] &= \mathbb{E}\left[\left(\sum_t \left(\beta_{it} - \tau_{gp}\right) D_{it}\right)^2\right] + \mathbb{E}\left[\left(\sum_t \left(\tau_{gp} - \beta\right) D_{it}\right)^2\right] \\
&\quad + 2\mathbb{E}\left[\left(\sum_t \left(\beta_{it} - \tau_{gp}\right) D_{it}\right)\left(\sum_t \left(\tau_{gp} - \beta\right) D_{it}\right)\right]
\end{aligned}$$

Since $\mathbb{E}\left[\left(\sum_t \left(\beta_{it} - \tau_{gp}\right) D_{it}\right) \mid \left(\sum_t \left(\tau_{gp} - \beta\right) D_{it}\right)\right] = 0$, under fixed D,

$$\begin{aligned}
\mathbb{E}\left[\left(\sum_t \left((\beta_{it} - \tau_{gp}) + (\tau_{gp} - \beta)\right) D_{it}\right)^2\right] &= \mathbb{E}\left[\left(\sum_t \left(\beta_{it} - \tau_{gp}\right) D_{it}\right)^2\right] + \left(\sum_t \left(\tau_{gp} - \beta\right) D_{it}\right)^2 \\
&\geq \mathbb{E}\left[\left(\sum_t \left(\beta_{it} - \tau_{gp}\right) D_{it}\right)^2\right]
\end{aligned}$$

so it is conservative for the target variance. □

Proof of Proposition 2. Part (i). Setting $\tilde{\tau}_g = \hat{\beta}$ for all g in the BJS formula gives treated residuals $r_{it} = Y_{it} - \hat{\alpha}_g - \hat{\gamma}_t - \hat{\beta}$ for $(i,t) \in \Omega_1$. The resulting variance estimator is $\hat{V} = \sum_i (\sum_t w_{it} \, r_{it})^2$, which coincides with the 2SDD GMM sandwich because the moment condition $g_i = \sum_t D_{it}(\tilde{Y}_{it} - \tilde{X}_{it}'\hat{\gamma} - \hat{\beta} D_{it})$ reduces to $\sum_t w_{it}(Y_{it} - \hat{\alpha}_g - \hat{\gamma}_t - \hat{\beta})$ after substituting the 2SDD first-stage estimates.

Part (ii). The Wooldridge ETWFE regression saturates treatment dummies at the cohort-time level, including one indicator D_{it}^{gt} per distinct cohort-time cell (g,t) with $g \in \{1,\dots,G\}$ and $t \in \{1,\dots,T\}$. The regression is

$$Y_{it} = X_{it}'\gamma + \sum_{g,t} \tau_{g,t} D_{it}^{gt} + u_{it},$$

so the fitted treatment effect for observation $(i,t) \in \Omega_1$ is $\hat{\tau}_{g(i),t}$, constant within each cohort-time cell. The full regression residual for a treated observation is therefore

$$\hat{u}_{it} = Y_{it} - X_{it}'\hat{\gamma} - \hat{\tau}_{g(i),t} = Y_{it} - \hat{\alpha}_g - \hat{\beta}_t - \hat{\tau}_{g(i),t}, \quad (i,t) \in \Omega_1.$$

Since the point estimators are algebraically identical, the weights w_{it} coincide across all three procedures. The Wooldridge sandwich variance is therefore

$$\hat{\sigma}_W^2 = \sum_i \left(\sum_t w_{it}\hat{u}_{it} \right)^2.$$

It remains to show this coincides with the BJS estimator under the cohort-time saturated auxiliary model $\tilde{\tau}_{it} = \hat{\tau}_{g(i),t}$. Under this auxiliary model, the BJS treated residuals are

$$r_{it} = Y_{it} - \hat{\alpha}_g - \hat{\beta}_t - \hat{\tau}_{g(i),t} = \hat{u}_{it}, \quad (i,t) \in \Omega_1,$$

where the last equality follows from the expression for $\hat{u}_{it}$ derived above. Substituting into the BJS variance formula gives

$$\hat{\sigma}_{\text{BJS}}^2 = \sum_i \left(\sum_t w_{it}\tilde{\varepsilon}_{it} \right)^2 = \sum_i \left(\sum_t w_{it}\hat{u}_{it} \right)^2,$$

confirming that the Wooldridge sandwich is the BJS estimator under the cohort-time saturated auxiliary model. □

B Additional Theoretical Results

B.1 Algebraic Equivalence

We show that 2SDD, BJS, and the Wooldridge ETWFE regression are algebraically identical as point estimators whenever they share the same nuisance specification, donor pool, and aggregation target. The equivalence is purely algebraic and requires no causal assumptions.

The saturated treatment design matrix is denoted H, with rows indexed by $(i,t) \in \Omega$ and columns indexed by cohort-time cells (g,s) for which some unit in cohort g is treated at period s. The (i,t)-th row of H is the one-hot indicator on column (G_i,t) when $D_{it} = 1$, and zero otherwise. We partition H by row into H_0 (rows in $\Omega_0(D)$) and H_1 (rows in $\Omega_1(D)$), and analogously partition X, Y into X_0, X_1, Y_0, Y_1.

Assumption B.1 (Rank)**.** *$X_0'X_0$ is invertible.*

Assumption B.2 (Common implementation)**.** *The 2SDD, BJS, and Wooldridge estimators share:*

(i) the same nuisance regressor matrix X, with columns spanning a common space;

(ii) the same donor pool $\Omega_0(D) \subseteq \{(i,t) : D_{it} = 0\}$ used to estimate the first-stage coefficient $\hat{\gamma}$;

(iii) the same aggregation weight vector $w \in \mathbb{R}^{N_1}$ with $\mathbf{1}'w = 1$, which is measurable with respect to the cohort-time partition of $\Omega_1(D)$: $w_{it} = w_{jt}$ whenever $G_i = G_j$ and $(i,t), (j,t) \in \Omega_1(D)$.

Assumption B.3 (Saturated treatment span)**.** *In the Wooldridge ETWFE/POLS regression $Y = X\gamma + H\beta + u$:*

(i) $H_0 = 0$ (treatment regressors are zero on untreated rows);

(ii) the column space of H_1 equals the span of the cohort-time indicator matrix on $\Omega_1(D)$, with one indicator per distinct cell (g,t) such that some unit in cohort g is treated at period t, and no further restrictions on β;

(iii) the column space of X_1 is contained in the column space of H_1.

Assumption B.3(ii) holds whenever Wooldridge's regression saturates the treatment dummies. If H imposes restrictions on β (e.g., homogeneous treatment effects across cohorts), the Wooldridge estimator can differ from the imputation estimator (6) and equivalence fails. Assumption B.3(iii) holds whenever X consists of unit and time fixed effects, or coarsenings thereof such as cohort and time indicators: on $\Omega_1(D)$, each unit indicator is a sum of cohort-time indicators for that unit, and each time indicator is a sum of cohort-time indicators across cohorts treated at that period. The condition can fail when X contains covariates that vary within cohort-time cells, in which case the three estimators may differ as point estimators.

Theorem B.1 (Algebraic equivalence of the three point estimators)**.** *Under Assumptions B.1–B.3,*

$$\hat{\theta}^{\text{2SDD}} = \hat{\theta}^{\text{BJS}} = \hat{\theta}^{\text{W}} = w'(Y_1 - X_1\hat{\gamma}), \qquad \hat{\gamma} = (X_0'X_0)^{-1}X_0'Y_0.$$

Proof. BJS. The BJS imputation estimator fits the untreated-outcome model on $\Omega_0(D)$ and imputes counterfactuals for treated observations: $\hat{\theta}^{\text{BJS}} = w'(Y_1 - X_1\hat{\gamma})$.

2SDD. The first stage estimates $\hat{\gamma}$ by OLS on $\Omega_0(D)$. The second stage forms residualised treated outcomes $\tilde{Y}_1 := Y_1 - X_1\hat{\gamma}$ and aggregates: $\hat{\theta}^{\text{2SDD}} = w'\tilde{Y}_1 = w'(Y_1 - X_1\hat{\gamma})$.

Wooldridge. The saturated regression is $Y = X\gamma + H\tau + u$, with normal equations

$$\begin{pmatrix} X'X & X'H \\ H'X & H'H \end{pmatrix} \begin{pmatrix} \hat{\gamma} \\ \hat{\tau} \end{pmatrix} = \begin{pmatrix} X'Y \\ H'Y \end{pmatrix}.$$

By Assumption B.3(i), $H_0 = 0$, so $X'H = X_1'H_1$, $H'H = H_1'H_1$, and $H'Y = H_1'Y_1$.

Step 1: $\hat{\gamma} = (X_0'X_0)^{-1}X_0'Y_0$. The second block of the normal equations gives

$$\hat{\tau} = (H_1'H_1)^{-1}H_1'(Y_1 - X_1\hat{\gamma}).$$

Substituting into the first block,

$$X'X\hat{\gamma} + X_1'H_1(H_1'H_1)^{-1}H_1'(Y_1 - X_1\hat{\gamma}) = X'Y.$$

Using $X'X = X_0'X_0 + X_1'X_1$, $X'Y = X_0'Y_0 + X_1'Y_1$, and writing $M_{H_1} := I - H_1(H_1'H_1)^{-1}H_1'$, this rearranges to

$$(X_0'X_0 + X_1'M_{H_1}X_1)\,\hat{\gamma} = X_0'Y_0 + X_1'M_{H_1}Y_1.$$

By Assumption B.3(iii), $\text{col}(X_1) \subseteq \text{col}(H_1)$, hence $M_{H_1}X_1 = 0$ and $X_1'M_{H_1} = 0$. The equation collapses to

$$X_0'X_0\,\hat{\gamma} = X_0'Y_0,$$

and Assumption B.1 gives $\hat{\gamma} = (X_0'X_0)^{-1}X_0'Y_0$.

Step 2: $w'\hat{\tau} = w'(Y_1 - X_1\hat{\gamma})$. By Assumption B.3(ii), $H_1(H_1'H_1)^{-1}H_1'$ is the orthogonal projection onto cohort-time cell-constant vectors on $\Omega_1(D)$. By Assumption B.2(iii), w is cohort-time measurable, so $w \in \text{col}(H_1)$ and $H_1(H_1'H_1)^{-1}H_1'w = w$. Combining with $\hat{\tau} = (H_1'H_1)^{-1}H_1'(Y_1 - X_1\hat{\gamma})$ from Step 1,

$$w'\hat{\tau} = w'H_1(H_1'H_1)^{-1}H_1'(Y_1 - X_1\hat{\gamma}) = w'(Y_1 - X_1\hat{\gamma}),$$

using self-adjointness of the projection in the first equality and $H_1(H_1'H_1)^{-1}H_1'w = w$ in the second.

Combining Steps 1 and 2 gives $\widehat{\theta}^{\mathrm{W}} = w'(Y_1 - X_1\hat{\gamma})$, matching the BJS and 2SDD expressions. $\square$

B.2 Comparison with TWFE in 2×2 case

We have two treatment groups indexed by $B \in \{0,1\}$ for the treated and untreated group respectively, and two time periods indexed $T \in \{0,1\}$ for pre and post periods respectively. The treatment variable is $D_{it} = B_{it}T_{it}$, so a unit needs to be both in the treatment group $B_{it} = 1$ and in the post period to be treated. Then, we can write the regression equation as:

$$Y_{it} = \beta_0 + \beta_1 B_{it} + \beta_2 T_{it} + \beta_3 D_{it} + \varepsilon_{it}.$$

We focus on the estimand and hence use the actual residual instead of the estimated residual i.e., ε_{it} instead of $\hat{\varepsilon}_{it}$. Let $\tilde{D}$ denote the treatment D with group and time fixed effects (B,T) partialled out. The estimands are:

$$\begin{aligned}
V_{TWFE} &= \mathbb{E}\left[\sum_t \tilde{D}_{it}^2\right]^{-1} \mathbb{E}\left[\left(\sum_t \tilde{D}_{it}\varepsilon_{it}\right)^2\right] \mathbb{E}\left[\sum_t \tilde{D}_{it}^2\right]^{-1} \\
V_{2SDD} &= \mathbb{E}\left[\sum_t D_{it}\right]^{-1} \mathbb{E}\left[\left(g_i + G_\gamma \psi_i\right)^2\right] \mathbb{E}\left[\sum_t D_{it}\right]^{-1}
\end{aligned}$$

Lemma B.1. $V_{TWFE} = V_{2SDD}$.

Proof. We characterize the estimands entirely in terms of primitive parameters, where b indexes the treatment group and t indexes time. Define $\mu := \mathbb{E}\left[B_{it}\right]$ and $v_{bt} := Var\left(Y_{it} \mid b, t\right) = \mathbb{E}\left[\varepsilon_{it}^2 \mid b, t\right]$. We show that: $V_{TWFE} = \frac{1}{\mu}\left(v_{11} + v_{10}\right) + \frac{1}{(1-\mu)}\left(v_{01} + v_{00}\right) = V_{2SDD}$.

We start with V_{TWFE}. To construct $\tilde{D}_{it}$, we decompose D_{it} as $D_{it} = \alpha_0 + \alpha_1 B_{it} + \alpha_2 T_{it} + e_{it}$, and express $\alpha_0, \alpha_1, \alpha_2$ in terms of primitives of the model by using moment conditions $\mathbb{E}\left[e_{it}\right] = \mathbb{E}\left[B_{it}e_{it}\right] = \mathbb{E}\left[T_{it}e_{it}\right] = 0$. We can use the results that $\mathbb{E}\left[D_{it}\right] = \mathbb{E}\left[D_{it}^2\right] = \frac{1}{2}\mu$, $\mathbb{E}\left[B_{it}\right] = \mu$, $\mathbb{E}\left[T_{it}\right] = 1/2$ and $\mathbb{E}\left[B_{it}D_{it}\right] = \mathbb{E}\left[T_{it}D_{it}\right] = \frac{1}{2}\mu$. The moment conditions yield:

$$\begin{aligned}
\mathbb{E}\left[D_{it}\right] &= \alpha_0 + \alpha_1\mu + \alpha_2\frac{1}{2} = \frac{1}{2}\mu \\
\mathbb{E}\left[B_{it}e_{it}\right] &= \mathbb{E}\left[B_{it}\left(D_{it} - \alpha_0 - \alpha_1 B_{it} - \alpha_2 T_{it}\right)\right] \\
&= \frac{1}{2}\mu - \alpha_0\mu - \alpha_1\mu - \alpha_2\frac{1}{2}\mu = 0
\end{aligned}$$

$$
\begin{aligned}
\mathbb{E}\left[T_{it}e_{it}\right] &= \mathbb{E}\left[T_{it}\left(D_{it}-\alpha_0-\alpha_1 B_{it}-\alpha_2 T_{it}\right)\right] \\
&= \frac{1}{2}\mu-\alpha_0\frac{1}{2}-\alpha_1\frac{1}{2}\mu-\alpha_2\frac{1}{2}=0
\end{aligned}
$$

Solving for the α's, we obtain $\alpha_0=-\frac{1}{2}\mu$, $\alpha_1=\frac{1}{2}$ and $\alpha_2=\mu$, so

$$
\tilde{D}_{it}=D_{it}+\frac{1}{2}\mu-\frac{1}{2}B_{it}-\mu T_{it}.
$$

By splitting the denominator into a component with $t=0$ and another with $t=1$,

$$
\begin{aligned}
\mathbb{E}\left[\tilde{D}_{i0}^2\right] &= \mathbb{E}\left[\frac{1}{2}\mu\left(D_{i0}+\frac{1}{2}\mu-\frac{1}{2}B_{i0}-\mu T_{i0}\right)\right]-\mathbb{E}\left[\frac{1}{2}B_{i0}\left(D_{i0}+\frac{1}{2}\mu-\frac{1}{2}B_{i0}-\mu T_{i0}\right)\right] \\
&= \frac{1}{4}\mu\left(1-\mu\right)
\end{aligned}
$$

and $\mathbb{E}\left[\tilde{D}_{i1}^2\right]=\frac{1}{4}\mu\left(1-\mu\right)$, so $\mathbb{E}\left[\sum_t \tilde{D}_{it}^2\right]=\frac{1}{2}\mu\left(1-\mu\right)$. Proceeding with the numerator, we can expand the expression:

$$
\mathbb{E}\left[\left(\sum_t \tilde{D}_{it}\varepsilon_{it}\right)^2\right]=\mathbb{E}\left[\tilde{D}_{i0}^2\varepsilon_{i0}^2+2\tilde{D}_{i0}\varepsilon_{i0}\tilde{D}_{i1}\varepsilon_{i1}+\tilde{D}_{i1}^2\varepsilon_{i1}^2\right].
$$

We have: $\mathbb{E}\left[\tilde{D}_{i0}\varepsilon_{i0}\tilde{D}_{i1}\varepsilon_{i1}\right]=0$ using the assumption that $\mathbb{E}\left[\varepsilon_{it}\mid g,t\right]=0$. Using the moment results that $\mathbb{E}\left[D_{i0}\varepsilon_{i0}^2\right]=0$, $\mathbb{E}\left[B_{i0}\varepsilon_{i0}^2\right]=\mu v_{10}$, $\mathbb{E}\left[T_{i0}\varepsilon_{i0}^2\right]=0$, $\mathbb{E}\left[\varepsilon_{i0}^2\right]=\mu v_{10}+\left(1-\mu\right)v_{00}$, $\mathbb{E}\left[D_{i1}\varepsilon_{i1}^2\right]=\mu v_{11}$, $\mathbb{E}\left[B_{i1}\varepsilon_{i1}^2\right]=\mu v_{11}$, $\mathbb{E}\left[T_{i1}\varepsilon_{i1}^2\right]=\mu v_{11}+\left(1-\mu\right)v_{01}$, and $\mathbb{E}\left[\varepsilon_{i1}^2\right]=\mu v_{11}+\left(1-\mu\right)v_{01}$, we obtain

$$
\begin{aligned}
\mathbb{E}\left[\tilde{D}_{i0}^2\varepsilon_{i0}^2\right] &= \mathbb{E}\left[\left(D_{i0}+\frac{1}{2}\mu-\frac{1}{2}B_{i0}-\mu T_{i0}\right)^2\varepsilon_{i0}^2\right] \\
&= \frac{1}{4}\mu\left(1-\mu\right)\left(\mu v_{00}+\left(1-\mu\right)v_{10}\right)
\end{aligned}
$$

and similarly, $\mathbb{E}\left[\tilde{D}_{i1}^2\varepsilon_{i1}^2\right]=\frac{1}{4}\mu\left(1-\mu\right)\left(\mu v_{01}+\left(1-\mu\right)v_{11}\right)$. Substituting these expressions back into the variance estimand yields the result.

Turning to the 2SDD variance, its components are:

$$
\begin{aligned}
G_\gamma &= -\mathbb{E}\left[\sum_t D_{it}\tilde{X}_{it}\right] \\
\psi_i &= \mathbb{E}\left[\tilde{X}_{0i}'\tilde{X}_{0i}\right]^{-1}\left(\tilde{X}_{0i}'\left(\tilde{Y}_{0i}-\tilde{X}_{0i}\gamma\right)\right)
\end{aligned}
$$

Here, $X = T$, $\tilde{Y}_{it} = Y_{it} - Y_{i0}$ and $\tilde{X}_{it} = X_{it} - X_{i0}$. We also have

$$\begin{aligned}
Y_{i1} - Y_{i0} &= \beta_0 + \beta_1 B_{i1} + \beta_2 T_{i1} + \beta_3 D_{i1} + \varepsilon_{i1} - (\beta_0 + \beta_1 B_{i0} + \varepsilon_{i0}) \\
&= \beta_2 + \beta_3 D_{i1} + \varepsilon_{i1} - \varepsilon_{i0}.
\end{aligned}$$

In this context, $\tilde{X}_{0i}$ only contains T because the constant and group FE have already been partialled out, so

$$\tilde{X}_{0i} = \begin{bmatrix} (T_{i0} - T_{i0}(1 - D_{i0}))(1 - D_{i0}) \\ (T_{i1} - T_{i0}(1 - D_{i0}))(1 - D_{i1}) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 - D_{i1} \end{bmatrix}$$

$$\tilde{Y}_{0i} = \begin{bmatrix} (Y_{i0} - Y_{i0})(1 - D_{i0}) \\ (Y_{i1} - Y_{i0})(1 - D_{i1}) \end{bmatrix} = \begin{bmatrix} 0 \\ (\beta_2 + \varepsilon_{i1} - \varepsilon_{i0})(1 - D_{i1}) \end{bmatrix}.$$

With these expressions, $\gamma = \mathbb{E}\left[\tilde{X}_{0i}'\tilde{X}_{0i}\right]^{-1}\mathbb{E}\left[\tilde{X}_{0i}'\tilde{Y}_{0i}\right] = \beta_2$, so

$$\begin{aligned}
\tilde{X}_{0i}'\left(\tilde{Y}_{0i} - \tilde{X}_{0i}\gamma\right) &= (1 - D_{i1})\left((\beta_2 + \varepsilon_{i1} - \varepsilon_{i0})(1 - D_{i1}) - (1 - D_{i1})\beta_2\right) \\
&= (\varepsilon_{i1} - \varepsilon_{i0})(1 - D_{i1})
\end{aligned}$$

Hence,

$$\begin{aligned}
\psi_i &= \mathbb{E}\left[1 - D_{i1}\right]^{-1}\left((\varepsilon_{i1} - \varepsilon_{i0})(1 - D_{i1})\right) \\
&= \frac{1}{1 - \mu}\left((\varepsilon_{i1} - \varepsilon_{i0})(1 - D_{i1})\right), \text{ and}
\end{aligned}$$

$$\mathbb{E}\left[\sum_t D_{it}\tilde{X}_{it}\right] = \mathbb{E}\left[D_{i1}\tilde{X}_{i1}\right] = \mu,$$

so $G_\gamma = -\mathbb{E}\left[\sum_t D_{it}\tilde{X}_{it}\right] = -\mu$. We finally have:

$$\begin{aligned}
g_i &= \sum_t D_{it}\left(\tilde{Y}_{it} - \tilde{X}_{it}'\gamma - \beta_3 D_{it}\right) \\
&= D_{i1}\left(\tilde{Y}_{i1} - \tilde{X}_{i1}'\beta_2 - D_{i1}\beta_3\right) = D_{i1}(\varepsilon_{i1} - \varepsilon_{i0})
\end{aligned}$$

and

$$\mathbb{E}\left[\left(g_i + G_\gamma \psi_i\right)^2\right] = \mathbb{E}\left[\left(D_{i1}\left(\varepsilon_{i1} - \varepsilon_{i0}\right) - \mu \frac{1}{1-\mu}\left(\left(\varepsilon_{i1} - \varepsilon_{i0}\right)\left(1 - D_{i1}\right)\right)\right)^2\right]$$

$$= \mu\left(v_{11} + v_{10}\right) + \frac{\mu^2}{(1-\mu)}\left(v_{01} + v_{00}\right)$$

Substituting these components into the variance expression yields the result. □

C Stata syntax

Suppose that y refers to the outcome, year the year, id the group, and d treatment status. In the simplest case, the two-stage difference-in-differences estimator can be obtained, along with valid cluster-robust asymptotic standard errors, via GMM using the single Stata command:

```
gmm (eq1: (y - {xb: i.year} - {xg: ibn.id})*(1-d)) ///
  (eq2: y - {xb:} - {xg:} - {delta}*d), ///
  instruments(eq1: i.year ibn.id) ///
  instruments(eq2: d, noconstant) winitial(identity) ///
  onestep quickderivatives vce(cluster id)
```

The did2s package (Butts, 2021) implements the same procedure more efficiently and scales more easily with individual fixed effects. The analogous implementation is as follows.

```
did2s y, first_stage(i.year ibn.id) second_stage(d) treatment(d) cluster(id)
```

Variations on the two-stage estimator (such as the two-stage event-study estimator) can be obtained using similar syntax, which we briefly mention below. We present both the gmm and did2s implementations.

Event Studies. We let wr be an indicator for having r periods since treatment. Let d0 denote the untreated observations.

```
gmm (eq1: (y - {xb: i.year} - {xg: ibn.id})*d0) ///
  (eq2: y - {xb:} - {xg:} - {b1}*w1 ... - {bR}*wR) ///
  instruments(eq1: i.year ibn.id) ///
  instruments(eq2: w1 ... wR, noconstant) winitial(identity) ///
  onestep quickderivatives vce(cluster id)
```

```
did2s y, first_stage(i.year ibn.id) second_stage(w1... w5) treatment(d) cluster(id)
```

The coefficients {b1} to {bR} are then our event study coefficients. 2SDD is a special case of the above where we only use b1. Design-based analysis does not change the code — it only changes the interpretation.

Parallel trends failure. If we want to estimate the first stage only using untreated data within a R periods of being treated, then we would define d0 accordingly.

```
gen d0 = (year - treatment_year > -R)*(1-d)
```

For did2s, use the following.

```
gen md0 = 1 - d0
did2s y, first_stage(i.year ibn.id) second_stage(d) treatment(md0) cluster(id)
```

Triple differences. With state, time, and subgroup as described in our main text, we can use the same `gmm` command as before, just that `eq1` is modified to be the following.

```
(eq1: (y - {xb: i.state##i.time i.state##i.subgroup i.time##i.subgroup})*d0)
```

Continuous and multivalued treatments. We can allow `d` to be continuous or multivalued without changing the syntax. For instance, the original 2SDD implementation would use the following. With a multivalued d, we need a constant when approximating the causal response function for treated units.

```
gen d0 = d==0
gmm (eq1: (y - {xb: i.year} - {xg: ibn.id})*d0) ///
  (eq2: y - {xb:} - {xg:} - {delta}*d), ///
  instruments(eq1: i.year ibn.id) ///
  instruments(eq2: d) winitial(identity) ///
  onestep quickderivatives vce(cluster id)
```

```
gen md0 = 1 - d0
did2s y, first_stage(i.year ibn.id) second_stage(contX) treatment(md0) cluster(id)
```

Reversible treatment and several treatments. We write our approach for several treatments as coding for the treatment path with reversible treatments is analogous. Suppose we have two binary treatments `d1` and `d2`.

```
gen d0 = d1==0 & d2==0
gen p1 = d1*(1-d2)
gen p2 = d2*(1-d1)
gen p3 = d1*d2
gmm (eq1: (y - {xb: i.year} - {xg: ibn.id})*d0) ///
  (eq2: y - {xb:} - {xg:} - {b1}*p1- {b2}*p2 - {b3}*p3), ///
  instruments(eq1: i.year ibn.id) ///
  instruments(eq2: p1 p2 p3, noconstant) winitial(identity) ///
  onestep quickderivatives vce(cluster id)
```

```
did2s y, first_stage(i.year ibn.id) second_stage(p1 p2 p3) treatment(d) cluster(id)
```

Test for treatment effect heterogeneity. We can conduct the joint test of whether interactions of treatment with covariates are significant. Suppose we have a `region` covariate.

```
gmm (eq1: (y - {xb: i.year} - {xg: ibn.id})*(1-d)) ///
  (eq2: y - {xb:} - {xg:} - {xf: region#d d}) ///
  instruments(eq1: i.year ibn.id) ///
  instruments(eq2: region#d d, noconstant) winitial(identity) ///
  onestep quickderivatives vce(cluster id)
test _b[xf:0b.region#1.d] == _b[xf:2.region#1.d] == 0

did2s y, first_stage(i.year ibn.id) second_stage(region#d d) treatment(d) cluster(id)
test _b[0b.region#1.d] == _b[2.region#1.d] == 0
```

D Regression-based difference-in-differences estimands

D.1 The TWFE estimand

From Equation (1), we can write

$$Y_{it} = \lambda_{g(i)} + \alpha_{p(t)} + \sum_{h=1}^{G}\sum_{q=h}^{P} \beta_{hq} 1(h,q)_{it} + e_{it}, \tag{9}$$

where $1(h,q)_{it}$ is an indicator for whether observation (i,t) corresponds to group h and period q, and $\mathbb{E}\left[e_{gpit} \mid g, p, (1(h,q)_{it})\right] = 0$.

Let $\tilde{D}_{it}$ denote the residual from a population regression of D_{it} on group and period fixed effects. By the Frisch-Waugh-Lovell theorem, the coefficient on D_{it} from a population regression of Y_{it} on D_{it} and group and period effects is

$$\begin{aligned}
\beta^* &= \frac{\mathbb{E}\left[\tilde{D}_{it} Y_{it})\right]}{\mathbb{E}\left[\tilde{D}_{it}^2\right]} \\
&= \frac{\mathbb{E}\left[\tilde{D}_{it} \sum_{h=1}^{G}\sum_{q=h}^{P} \beta_{hq} 1(h,q)_{it}\right]}{\mathbb{E}\left[\tilde{D}_{it}^2\right]} \\
&= \sum_{h=1}^{G}\sum_{q=h}^{P} \frac{\mathbb{E}\left[\tilde{D}_{it} 1(h,q)_{it}\right]\beta_{hq}}{\mathbb{E}\left[\tilde{D}_{it}^2\right]} \\
&= \sum_{g=1}^{G}\sum_{p=g}^{P} \omega_{gp}\beta_{gp}.
\end{aligned}$$

where ω_{gp} is the coefficient from a regression of $1(h,q)_{it}$ on D_{it} and group and period fixed effects. The second equality uses the facts that e_{it} is mean-independent of the regressors and that $\tilde{D}_{it}$ is uncorrelated with group and period effects by construction.[26]

The weight ω_{gp} that difference in differences places on β_{gp} is the coefficient on D_{it} from a regression of $1(g,p)_{it}$ on D_{it} and group and period fixed effects. By the Frisch-Waugh-Lovell theorem, this is equivalent to the slope coefficient from a population regression of $1(g,p)_{it}$ on the residual from an auxiliary regression of D_{it} on group and period effects. Using the two-way within or double-demeaned transformation, this residual can be expressed as

$$\tilde{D}_{it} = [D_{it} - \Pr(D_{it} = 1 \mid g)] - [\Pr(D_{it} = 1 \mid p) - \Pr(D_{it} = 1)]. \tag{10}$$

Since $\mathbb{E}\left[\tilde{D}_{it}^2\right] = \mathbb{E}\left[\tilde{D}_{it} D_{it}\right]$, ω_{gp} can also be expressed as

$$\begin{aligned}
\omega_{gp} &= \frac{\mathbb{E}\left[1(g,p)_{it}\tilde{D}_{it}\right]}{\operatorname{Var}\left[\tilde{D}_{it}\right]} \\
&= \frac{\mathbb{E}\left[\tilde{D}_{it} \mid 1(g,p)_{it} = 1\right]\Pr\big(1(g,p)_{gpit} = 1\big)}{\mathbb{E}\left[\tilde{D}_{it} \mid D_{it} = 1\right]\Pr(D_{it} = 1)} \\
&= \frac{[1 - \Pr(D_{it} = 1 \mid g) - (\Pr(D_{it} = 1 \mid p) - \Pr(D_{it} = 1))]\Pr(g,p)}{\sum_{g'=1}^{G}\sum_{p'=g'}^{P}[1 - \Pr(D_{it} = 1 \mid g') - (\Pr(D_{it} = 1 \mid p') - \Pr(D_{it} = 1))]\Pr(g',p')},
\end{aligned} \tag{11}$$

where the final equality uses Equation (10).

D.2 The stacked difference-in-differences estimand

In the stacked approach, a new dataset is created for each treated group, containing observations on that group $\overline{R}$ periods before, and $\bar{P}$ periods after, the treatment is adopted, as well as on units that are not yet treated during these periods. These group-specific datasets are stacked, and outcomes are regressed on treatment status and dataset-specific group and period fixed effects:

$$Y_{cit} = \lambda_{cg(i)} + \alpha_{cp(t)} + \beta D_{cit} + \varepsilon_{it},$$

where cit indexes the (i,t)th observation of dataset c.

Let D_{cit} be an indicator for whether i is treated at time t in dataset c, and D_{rcit} be an indicator for whether i has been treated for $r \in \{1, \ldots, \bar{P}\}$ periods as of time t in dataset c. Let $\tau = \bar{P}/(\bar{P} + \overline{R} + 1)$ denote the fraction of periods during which treated units in any

[26]This, and the related result in Sun and Abraham (2021), can also be established by thinking of the term $\sum_{h=1}^{G}\sum_{q=h}^{P}\beta_{hq}1(h,q)_{it}$ in Equation (9) as an omitted variable, and taking its projection onto the included regressors.

group-specific dataset are treated, π_c denote the fraction of units in dataset c that belong to the treatment group, and ρ_c denote size of dataset c relative to the stacked dataset.

Extending the logic of the previous section, the weight ω_{rg} that stacked DD places on the r-period average treatment effect β_{rg} for group g is given by the slope coefficient from a population regression of D_{rcit} on the residual $\tilde{D}_{cit}$ from a regression of D_{cit} on dataset×period and dataset×group effects.[27] This residual is

$$\tilde{D}_{cit} = D_{cit} - P(D_{cit} = 1|g, c) - [P(D_{cit} = 1|p, c) - P(D_{cit} = 1|c)],$$

where statements conditional on c are true in the population corresponding to dataset c. Using this expression and adapting (11) to the stacked setting,

$$\begin{aligned}\omega_{rg} &= \frac{[1 - \tau - (\pi_c - \tau\pi_c)]P(D_{rcit} = 1)}{\sum_{c=1}^{G}\sum_{p=1}^{\bar{P}}[1 - \tau - (\pi_c - \tau\pi_c)]P(D_{rcit} = 1)} \\ &= \frac{(1-\tau)(1-\pi_c)\tau\pi_c\rho_c}{\sum_{c=1}^{G}\sum_{p=1}^{\bar{P}}(1-\tau)(1-\pi_c)\tau\pi_c\rho_c} \\ &= \frac{(1-\pi_c)\pi_c\rho_c}{\bar{P}\sum_{c=1}^{G}(1-\pi_c)\pi_c\rho_c}.\end{aligned}$$

E Alternative approaches to testing parallel trends

There are alternative approaches to testing the validity of parallel trends within the two-stage framework. BJS2024 recommend testing for parallel trends by including leads of treatment status in the first stage of the estimator, noting that their approach can, under some conditions, circumvent concerns about conditioning difference-in-differences estimates on passing tests for parallel trends (note that inference in this approach is based on standard OLS asymptotics). Another approach is to assume that parallel trends holds up to $\underline{R}^* + 1$ periods before the adoption of the treatment, then use the two-stage procedure to estimate the $\underline{R}^*$ pre-treatment placebo ATTs (i.e., the coefficients on D_{rit} for $r \in \{-\underline{R}^*, \dots, -1\}$). This approach is also suggested by Liu, Wang and Xu (2022), who develop an equivalence test to increase the power of tests based on this idea.[28]

[27] This is because the correct model can be expressed as

$$Y_{cit} = \lambda_{cg(i)} + \alpha_{cp(t)} + \sum_{c=1}^{C}\sum_{r=1}^{\bar{R}}\beta_{rc}D_{crit} + e_{cit},$$

where $\beta_{rc} = \beta_{rg}$ when $D_{crit} = 1$.

[28] While all of the methods discussed above are capable of identifying violations of parallel trends, none of them reliably identify parameters that can be interpreted as average deviations from trends in pre-treatment

The two-stage framework suggests another approach still, this one motivated by the fact that it is not necessary to use all pre-treatment periods to identify the group (or individual) and time effects used by the second stage of the estimator. For example, instead of using all untreated observations, the first stage can be estimated from the sample of all observations for never-treated units (from which the period effects and group effects for never-treated units are identified) as well as all observations for eventually-treated units in the period immediately before they adopt the treatment (from which the group effects for treated units are identified).[29] Under the normalization that parallel trends holds in the last pre-treatment period (i.e., that eventually-treated units experience the same time effects in that period as never-treated units), the coefficients on the D_{rit} for $r \in \{-\underline{R}, \ldots, -1\}$ for this variant of the two-stage procedure identify average pre-treatment deviations among eventually-treated units from never-treated units' trends.[30] The consistency of our procedure in this case follows from a simple modification of the theory developed in section 2.3.

Although this restriction of the first-stage sample may reduce the efficiency of the second-stage estimates, it addresses some of the challenges associated with interpreting coefficients that represent tests of parallel trends from within the two-stage framework (cf. footnote 28 and Roth 2024). In Figure A12, we show that two-stage estimates obtained using this modified procedure correctly identify both pre- and post-treatment trends in the setting where Roth (2024) shows that the default dCDH2020, CS2021, and BJS2024 estimators do not. While the coefficients on leads of treatment status from this modified procedure are more readily interpretable, the analogous coefficients from the "standard" two-stage approach (i.e., using the full untreated sample in the first stage) still represent valid tests of parallel trends, even if they cannot be interpreted as average deviations from never-treated trends. A further advantage of this modified procedure is that it may offer superior performance in cases when the divergence between untreated outcomes between eventually- and never-treated units increases over time (an advantage that dCDH2024 argue is shared by other estimators that do not compare treated observations to all untreated observations). Finally, we note that, at least in models without covariates, this variant of 2SDD is numerically equivalent to

periods. Second-stage coefficients on leads of treatment status test whether average first-stage residuals are close to zero in pre-treatment periods, first-stage coefficients on such leads presumably identify a (potentially non-convex) weighted average of deviations from trend for all groups and periods, and placebo ATTs only represent such deviations under the assumption that parallel trends holds prior to the adoption of the placebo treatment. This contrasts with traditional event-studies based on two-way fixed-effects regressions with homogeneous duration-specific average treatment effects, in which the coefficients on leads can be interpreted as average deviations from trends, subject to a normalization.

[29] The last treated cohort can be used as the never-treated cohort in the absence of a pure control group.

[30] The normalization required here is the same as that required for traditional two-way fixed-effects event studies.

the estimator developed in SA2021 and the never-treated variant of CS2021.[31]

F Discussion of assumptions and extensions

The simplicity of our regression-based estimation and inference procedure allows us to flexibly incorporate several extensions and relaxations of the assumptions. We provide sample `Stata` code in Appendix C both using the in-built `gmm` command and the `did2s` package.

Other average treatment effects. While the discussion so far has focused on identification of the overall and duration-specific ATTs, as these are the most common objects of interest in difference-in-difference analyses, the logic behind our two-stage approach also applies to other average treatment effect measures. In particular, it can be used to identify group×time-specific ATTs by amending the second stage to include a full set of group×time indicators, which can be examined individually or aggregated to form further summary measures of the treatment. Similarly, modifying the second stage to include full sets of group or time indicators identifies group- or time-specific ATTs.

Parallel trends assumption. The theoretical results presented above assume parallel trends for every group and between every pair of consecutive time periods, as in dCDH2020 and SA2021. This assumption is stronger than the one used in CS2021, who only require parallel trends between treated groups and never-treated groups after their treatment time. While this accommodates cases where parallel trend fails prior to treatment time, the distinction may not matter in practice, as testing for parallel trends prior to treatment time is often used as a proxy for the infeasible test for parallel trends post treatment. If the stronger version of parallel trends fails, researchers tend to have little confidence in the weaker version.

Nevertheless, our procedure can be modified to accommodate the weaker version. If we only believe in the weak version of parallel trends, our procedure can be modified to estimate

[31]The reason for this equivalence is that these estimators can be obtained from the regression model

$$Y_{it} = \lambda_{g(i)} + \gamma_t + \sum_{r\neq -1} \sum_{g=1}^{G} \beta_{rg} 1(G_i = g, W_{rit} = 1) + \varepsilon_{it}.$$

By the FWL theorem, the estimated $\lambda_{g(i)}$ and γ_t can be obtained by regressing Y_{it} on the residuals from a regression of cohort and time indicators on the $1(G_i = g, W_{rit} = 1)$. This auxiliary regression perfectly predicts cohort membership and time for observations on eventually treated units that do not correspond to the period immediately preceding adoption, so the residuals for these observations are zero. In a model without a constant, this implies that the cohort and time fixed effects are estimated from the subsample corresponding to never-treated units and the period preceding adoption for treated units, and is therefore equivalent to regressing $Y_{it} - \hat{\lambda}_{g(i)} - \hat{\gamma}_t$ on the $1(G_i = g, W_{rit} = 1)$, where $\hat{\lambda}_{g(i)}$ and $\hat{\gamma}_t$ are first-stage estimates obtained using this subsample.

the first stage using only never-treated observations and the first pre-treatment period for eventually-treated units (which is necessary in order to estimate the unit FEs for treated units). In contrast, even if the stronger version of parallel trend holds, the CS2021 approach does not yield a more precise estimate because it does not make use of data from treated observations before relative time -1.

Our approach can be adapted to reduce bias when the parallel trends assumption is violated. The potential for a larger bias arises if the parallel trends assumption does not hold exactly and the difference in trends between groups increases over time. To reduce the bias, we can simply estimate the first stage using untreated data within a few periods of being treated. Group-specific linear trends may also be included in the regression-based approach to remove the group trends directly. In particular, X_{it} in Equation (3) may include $1\{g(i) = g\}t$, where $g(i)$ denotes the group to which observation i belongs.

Triple differences. Triple differences-in-differences use the evolution of observed outcomes for auxiliary untreated units to control for potential violations of parallel trends, and can easily be accommodated from within our two-stage framework. For example, if only members of states that belong to a particular subgroup are treated, then the first-stage regression can be modified to include state, time, and state×time, state×subgroup and time×subgroup fixed effects, which can be collected into the vector X_{it}. Under the hypothesis that state-level deviations from parallel trends are identical across subgroups within states, two-stage estimates will recover the overall ATT.

Serial correlation. It is possible to adapt the procedure to obtain an efficient estimator even in the presence of serial correlation (Appendix B). Since the estimator is identical to the imputation estimator of BJS2024, it is known that the estimator is efficient in the canonical normal homoskedastic model. If there is serial correlation in the error term following an AR(1) process for each i, we can make a simple adjustment to the regression. If ε_{it} in Equation (3) is AR(1) with correlation parameter ρ, then Equation (3) can be written as $Y_{it} = \rho Y_{it-1} + \delta_{it} D_{it} + \tilde{\lambda}_i + \tilde{\gamma}_t + \nu_{it}$, with $\nu_{it} | D_{it}, Y_i^{t-1} \sim \mathcal{N}(0, \sigma^2)$ for an appropriately defined $\tilde{\lambda}_i, \tilde{\gamma}_t$ and $Y_i^{t-1} := \{Y_{i1}, Y_{i2}, \cdots, Y_{it-1}\}$. Our procedure can be analogously implemented by: (1) regressing Y_{it} on Y_{it-1}, X_{it}, and fixed effects for observations with $D_{it} = 0$, (2) regressing $Y_{it} - \hat{\rho} Y_{it-1} - (1 - \hat{\rho})\widehat{\lambda_{g(i)}} - \widehat{\alpha_t} + \hat{\rho}\hat{\alpha}_{t-1}$ on D_{it}. Since the OLS estimator coincides with the maximum likelihood estimator, the resulting estimator is efficient under the normal homoskedastic benchmark. While the estimated coefficients are not the coefficients of interest due to the Nickell bias, they can be combined to recover the coefficients of interest in a manner detailed in the appendix.[32]

Continuous and multivalued treatments. Our approach to estimation and inference extends to continuous treatments and discrete (non-binary) treatments. In this setting, observations have $D_{it} = 0$ prior to treatment, but the treatment value may be continuous post-treatment. The two-stage procedure still applies, with the first-stage consisting of a regression of the outcome on X for $D_{it} = 0$ to obtain $\hat{\gamma}$, and the second stage consisting of a regression of $Y_{it} - X_{it}'\hat{\gamma}$ on D_{it} to obtain $\hat{\beta}$. With a continuous treatment, the two-stage procedure identifies a (positive) Yitzhaki-weighted average of the derivatives of the causal response function (Yitzhaki, 1996; Angrist and Krueger, 1999; Angrist and Pischke, 2009). Inference proceeds through GMM as before.[33]

Anticipation effects. The procedure can be extended to accommodate anticipation effects. If the treatment is anticipated for r periods before adoption, we can redefine treated to mean having adopted the treatment for at least r periods.

Reversible treatment and several treatments. The procedure can be extended to accommodate reversible treatment and having several treatments. If the treatment is reversible, one way to apply our results is to use the (potentially strong) assumption that there are no within-unit spillovers of the treatment to future periods. Alternatively, if there are within-unit spillovers of the treatment to future periods, we can define W_{it} in Section 2.4 as a vector of indicators for the treatment path, which is defined as the sequence of treatment indicators since first treatment.[34] Observations that have been treated prior to t are excluded from the first stage. The asymptotics hold when there are many observations with the same treatment path. The estimands remain interpretable as the corresponding coefficients are effects relative to the untreated group. If there are several treatments, say D_1 and D_2, then we can similarly define each treatment path as a tuple of the treatment duration of (D_1, D_2), so W_{it} is a vector of indicators for every combination of these tuples. The rest of the procedure and interpretation are identical to that of having reversible treatment with within-unit spillovers.

[32]While Harmon (2024) emphasizes the efficiency advantages of alternative approaches under strongly correlated errors, the procedure outlined here highlights how our estimator can be adapted to such settings.

[33]When using this approach to approximate the causal response to a multivalued treatment, the second-stage regression should include a constant term.

[34]For instance, $(0, 1, 0, 1)$ and $(0, 1, 1, 1)$ are two different treatment paths. While both groups begin being treated in the second period, the only first group becomes untreated in the third period. The coefficient on the third and fourth periods are then allowed to be different for the two groups to accommodate the different treatment paths.

Design-based analysis. The 2SDD estimand is also interpretable in a design-based world. Let A_{1i} denote the number of periods that individual i has been treated, with $A_{1i} = 1$ in the period that i was first treated. Further, assume that $\beta_{it} = \beta_i$ for all t, which, unlike our base framework, is non-random. Define the estimand as $\beta := \mathbb{E}[\beta_{it} \mid D_{it} = 1]$. Using an argument similar to the proof of Lemma A.1 in Appendix A, $\beta = \text{plim}\left(\sum_{i=1}^N \sum_{t=1}^T D_{it}\right)^{-1}\left(\sum_{i=1}^N \sum_{t=1}^T D_{it}\beta_{it}\right)$. Hence, in the setting with staggered treatment adoption, $\beta = \text{plim}\left(\sum_{i=1}^N A_i\right)^{-1}\left(\sum_{i=1}^N A_i\beta_i\right)$. Under the Athey and Imbens (2022) setup where the adoption time of treatment is as good as random, A_i is randomly assigned across individuals in our setting, so $\frac{1}{N}\sum_{i=1}^N A_i \xrightarrow{p} a$, and $\mathbb{E}[A_i] = a$ for all i.[35] Thus, the estimand becomes $\beta = \frac{1}{aN}\mathbb{E}\left[\sum_{i=1}^N A_i\beta_i\right] = \frac{1}{aN}\sum_{i=1}^N \mathbb{E}[A_i]\beta_i = \frac{1}{N}\sum_{i=1}^N \beta_i$, which is exactly the average treatment effect (ATE). Inference is straightforward due to the GMM framework: it suffices to cluster on i since A_i is randomly assigned.

Linear combination of treatment effects. The procedure can also be extended to estimating any linear combination of coefficients. Recall that we have the model $Y_{it} = D_{it}\beta_{it} + X_{it}'\gamma + \varepsilon_{it}$ with $\mathbb{E}\left[\varepsilon_{it} \mid \{D_{it}, X_{it}\}_{t=1}^T\right] = 0$. This model implies that $\mathbb{E}[Y_{it} - D_{it}\beta_{it} - X_{it}'\gamma] = 0$. We are interested in $\tau := w_{it}\beta_{it}$, where w_{it} is a non-stochastic weight. Due to the moment condition, and w_{it} being non-stochastic,

$$\mathbb{E}[w_{it}Y_{it} - w_{it}X_{it}'\gamma] - \mathbb{E}[D_{it}]w_{it}\beta_{it} = 0.$$

Assume that heterogeneity in $\mathbb{E}[D_{it}]$ occurs at some level h, and ζ is the vector of values it can take, so that $\mathbb{E}[D_{it}] = 1(h)_{it}'\zeta$. Assume that ζ is either known or can be consistently estimated, and all elements of ζ are nonzero. Then, by summing $w_{it}\beta_{it} = \mathbb{E}[w_{it}Y_{it} - w_{it}X_{it}'\gamma]/\mathbb{E}[D_{it}]$ over i, t:

$$\tau = \sum_{i,t} w_{it}\beta_{it} = \sum_{i,t} w_{it}\mathbb{E}\left[\frac{Y_{it} - X_{it}'\gamma}{1(h)_{it}'\zeta}\right]$$

[35] The difference in identifying assumptions not only affects the interpretation of the estimand, but also affects inference. The design-based environment models the assignment process A_i instead of $Y_i(0)$, so parallel trends is no longer required to interpret β as the ATE. For inference in the design-based environment, if there is clustered assignment for A_i on dimension C where several individuals i can belong to some cluster c, then researchers should cluster at the c level instead of i. However, if we instead assume parallel trends in our original environment, even if there is clustered assignment, it suffices to cluster on unit i if we are targeting a conditional estimand (i.e., the ATT): by conditioning on D_{it}, the dependence structure is irrelevant for inference.

Hence, writing everything as a system of moment conditions,

$$\mathbb{E}\left[\begin{array}{c} 1(h)_{it}\left(D_{it}-1(h)_{it}'\zeta\right) \\ X_{it}\left(1-D_{it}\right)\left(Y_{it}-X_{it}'\gamma\right) \\ \tau-w_{it}\left(\frac{Y_{it}-X_{it}'\gamma}{1(h)_{it}'\zeta}\right) \end{array}\right]=0$$

The just-identified system of equations enables the application of GMM in the same way as before.

Test for treatment effect heterogeneity. The GMM approach also allows us to test for treatment effect heterogeneity. One approach is to use the fact that, under the null of constant treatment effects, the two-way fixed effects estimator has the same probability limit as the 2SDD estimator that is robust to heterogeneity. Then, we can test if the two estimators are equal. An alternative approach is to obtain coefficients in the second stage that correspond to group (or covariate-specific) treatment effects. We can then use a standard F test to jointly test if the coefficients are equal.